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Aymen BAHRI

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Simulation of Interior Permanent Magnet Synchronous Machine using EMWorks software
Comparing Toyota Prius Motor versus Tesla '3 Motor

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Mr. Riadh ABDELATI	**President**
Mr. Ramzi HADAJI	**Examiner**
Ms. Hanen BERRIRI	**Supervisor**
Mr. Majdi EL FAHEM	**Guest**

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Dedication

I would like to dedicate this work to my parents. for whom I am eternally grateful...

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Abstract

In this end of study report, we will be interested in 2D and 3D design, simulation, analysis and optimization of electric motors for the automotive industry, based on finite element method.

electromagnetic simulation Software used in the simulation are EMS and EMWorks2D that are Embedded into SolidWorks. The Interior Permanent Magnet Synchronous Motors (IPMSM) characterized with their high efficiency, high-power factor, high-power density and excellent flux weakening capabilities, are one of the best choices for use in electric vehicles, especially electric cars.

An analytical method has been developed based on d- and q- axis equivalent circuit model, which is used to determine the parameters impact on motor output torque.

Also, Finite Element Method (FEM) will be briefly presented to get an idea about how FEM software calculate field equations solutions.

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Nomenclature

ϵ	electric permitivity
λ_m	magnet flux linkage
$\lambda_{a,b,c}$	phases a,b,c flux linkage
μ	magnetic permeability
μ_{rec}	relative coil permeability
ω	angular velocity
ω_e	electric angular velocity velocity
ω_m	angular velocity
ρ	electric charge density
σ	electrical conductivity
σ^M	Maxwell stress tensor
θ_e	electric angle
θ_m	mechanical angle
$\vec{A}$	magnetic potential
$\vec{B}$	magnetic flux density
$\vec{E}$	electric field
$\vec{F}$	Lorentz force
$\vec{H}$	magnetic field
$\vec{J}$	current density
B_r	Remanent flux

D electric displacement

EV electric vehicle

H_c corecive force

HEV electric vehicle

i_d, i_q direct and quadrature axis currents

$i_{a,b,c}$ phase currents

$IPMSM$ Interior Permanent Magnet Synchronous Motor

kOe kilo-Orested

L_d, L_q Direct axis and quadrature axis inducatances

$L_{a,b,c}$ phase inductance

M_{CP} Clarke-Park matrix

$MGOe$ mega-gauss-oersted

Oe Oersted unit for magnetic field H in the CGS system

P pair pole number

$PMSM$ Permanent Magnet Synchronous Motor

R residual

T_e electric torque

T_m mechanical Torque

W_e magnetic energy

Introduction

Nowadays, the trend is toward the use of electric and hybrid electric cars over fears of a looming environmental disaster.
Global warming has brought the electric car and its close cousin, the hybrid back from obscurity and perhaps the biggest change in electric vehicles in the last few years has been the development of the lithium battery [2], which has a reasonable specific energy and a more reasonable charge time than previous batteries. This has ultimately led to a series of commercial vehicles such as the Toyota Prius, Chevrolet Volt, the Nissan Leaf and, the Tesla sport.

Indeed, the multiple advantages offered by electric cars are normally associated with benefits to the environment and saving energy. Those benefits include reducing local pollution from the cars themselves, reducing dependence on oil and other fossil fuels and reduction of carbon emissions.

In addition to the advancement of batteries, various types of electric motors have been investigated for use in electric and hybrid electric cars, Permanent magnet motor (PM motor) is the dominant drive motor for two main reasons:Firstly, field current needed for induction machine is not necessary, and secondary copper loss does not occur,therefore high efficiency can be achieved. Secondly, magnetic flux can be dense due to use of rare earth magnets, and thus the drive motor can be downsized[3].

The interior permanent magnet synchronous motor (IPMSM) is currently the most popular electric motor for EVs[4]. Hence, To respond to the demand for protecting the environment, the transition toward electric cars must be accelerated.This can be accomplished by modelling and simulation to make those systems reliable and robust.

Computer-based modelling and simulation has gradually been replacing the traditional, laborious hand-calculations, drafting, physical testing and model building approach to many design tasks and over the recent years, this technology has significantly expanded its capabilities, aided by more powerful processors, greater visualization capacities, more sophisticated software allowing new functionality across multiple and more widely diffused platforms. Modelling and simulation is conducted on a variety of technological platforms, including computer-aided design (CAD), using a number of techniques such as computational fluid dynamics (CFD) and finite element analysis (FEA) [5].

In this context, our project deals with 2D and 3D design, simulation, and optimization of electric motors for the automotive industry, based on finite element method electromagnetic simulation.

EMWorks provides great tools based on finite element method to simulate electromechanical systems, we will be using EMS and EMWorks2D to simulate two commercial electric motors which are the Tesla Model 3 2018 and the Toyota Prius 2004, and we will be comparing their performance. This will outline how effective simulation software and how can be used to understand and investigate complicated systems.

The report will be organized as follows:
In **Chapter 1** we will be briefly presenting the internship framework

In **Chapter 2** we will be presenting the different types of electric Motors used in electric cars , especially AC machines, a brief introduction of Finite element method for electromagnetism is illustrated so that it's worth understanding how FEA software products work and a steady state DQ analysis is presented in order to understand how we can get a maximum starting torque using field weakening technique.

In **Chapter 3** we will be comparing 3D and 2D finite element analysis, A surface mounted permanent magnet motor model is used, the study will be just comparing No load torque constraints by computation performance needs when doing a loaded analysis. After that we will be presenting the Toyota Prius 2004 motor specification and the Tesla Model 3 2018 motor specifications, then we will be simulation and comparing their performance. We will also change the geometry and materials of the Tesla motor and see their effects on performance.

For conclusion, chapter **Chapter 4** briefly summarizes the research in this thesis and addresses future works.

Chapter 1

Internship framework

1.1 Introduction

In this first chapter, the host company will be introduced. It is a company specializing in electrical CAD and FEA software production. EMWorks offers four software which are gold certified products from the French company Dassault systems. Also, the objectives of the internship will be presented.

1.2 Company overview

EMWorks is a leading provider of electromagnetic simulation software for electrical and electronics design, and it's founded in 2000 and it's located in Canada and Germany. EMWorks products can be used to analyise and optimize a wide range of products such as transformers, antennas, PCB devices, electric motors ....
Today, EMWorks products and services are widely used worldwide by both industry and academia.

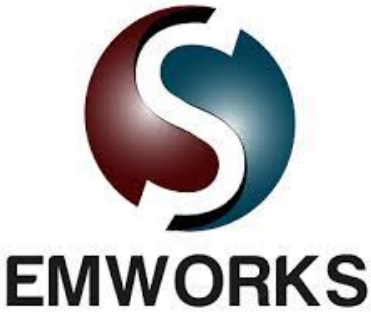


Figure 1.1: EMWorks Logo

there are more than 500 universities around the world are working with EMWorks Inc. tools. EMS & HFWorks have been great tools for students, researchers and teachers all over the world.

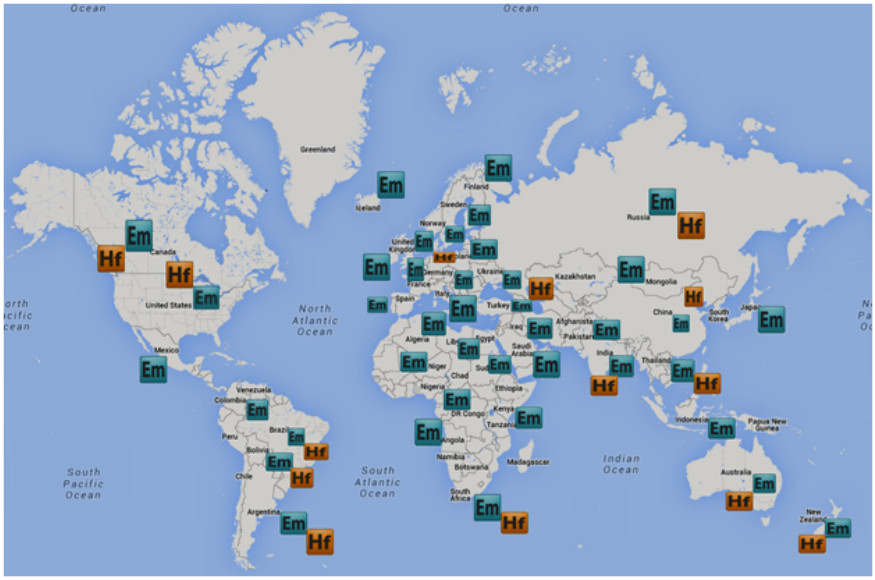


Figure 1.2: EMworks Products Map

University labs and teaching institutes using EMS and HFWorks software



Figure 1.3: Universities using EMS and HF works

EMWorks is currently servicing a list of global clients such as Ford, John Deere, AIRBUS...



(a) (b)

Figure 1.4: Featured Costumers and Clients

1.3 Company products

EMWorks' flagship products are **EMS**, **HFWorks**, **MotorWizard**, and **EMWorks2D**. All of those products are fully and seamlessly embedded in **SOLIDWORKS**. EMS is also embedded in Autodesk Inventor.

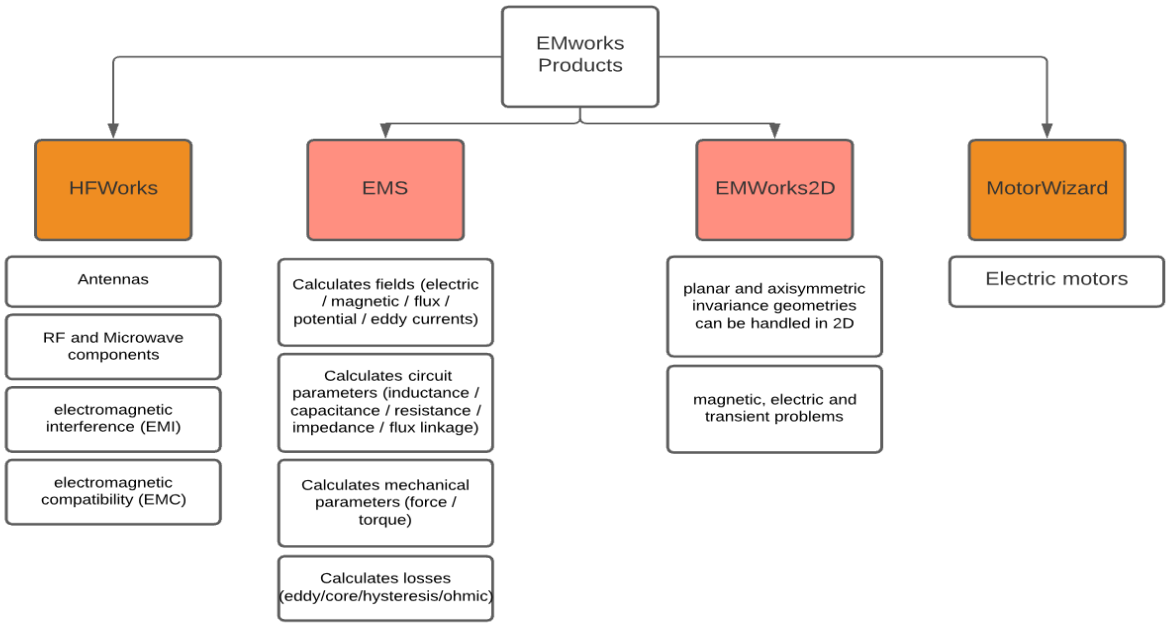


Figure 1.5: EMWorks products

In addition to handling frequencies from DC to millimeter wave, EMWorks products come with built-in integrated thermal, structural, and motion analyses in a hassle-free multiphysics platform.

1.3.1 HFWorks

HFWorks is an antenna and electromagnetic simulation software for RF, Microwave, mm-wave, and high speed digital circuits. HFWorks solves electromagnetic radiation, electromagnetic waves, electromagnetic propagation, electromagnetic resonance, electromagnetic interference (EMI), electromagnetic compatibility (EMC), and signal integrity (SI) problems for RF/MW frequencies and beyond. It uses state-of-the-art finite element solvers and meshing technologies to compute fields as well as antennas and circuit parameters. It can simulate single antenna elements as well as multiple array antenna configurations. HFWorks can also be used for time domain computations such as TDR and Eye Diagram. It can predict power handling capabilities of 3D structures and localize potential field breakdown areas. It also provides the capability to simulate RF microwave heating as a function of applied power.



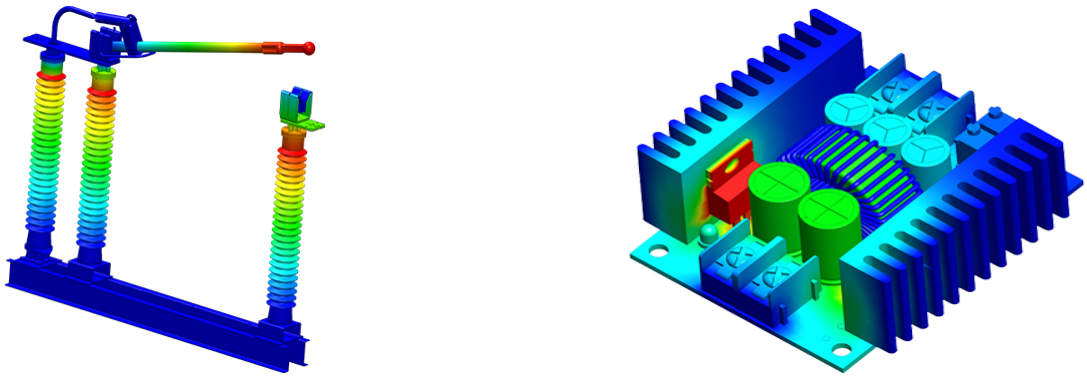
(a) Printed-Filter (b) Rat-Race-coupler

Figure 1.6: HFworks applications examples

1.3.2 EMS

EMS is an electromagnetic field simulation software which calculates fields (electric / magnetic / flux / potential / eddy currents), circuit parameters (inductance / capacitance / resistance / impedance / flux linkage), mechanical parameters (force / torque), and losses (eddy/core/hysteresis/ohmic). EMS is a Gold Certified Add-in to SolidWorks and an Add-in to Autodesk Inventor.It which enables as to simulate the most intricate electrical machines, motors, generators, sensors, transformers, high voltage apparatus, high power machines, electrical switches, valves, actuators, PCB's, levitation machines, loudspeakers, permanent

magnet machines, NDT equipment, inverters, converters, bus bars, inductors, bushings, or biomedical equipment.

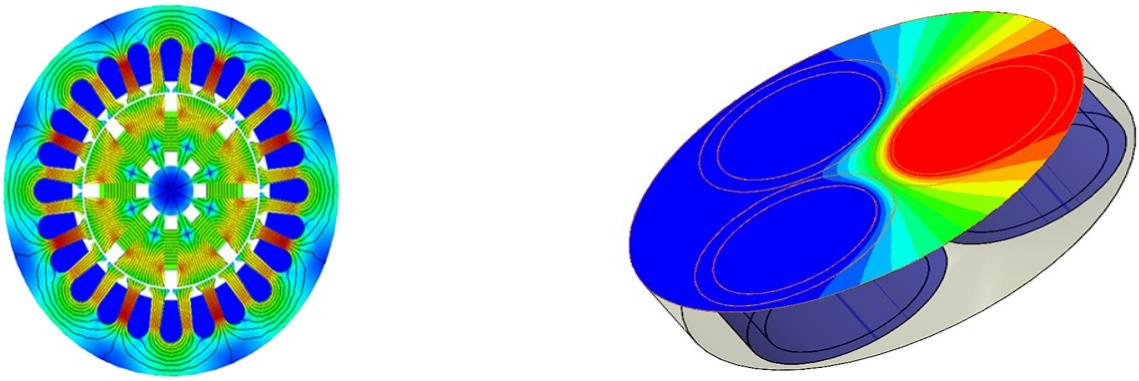


(a) High-Voltage-Switcher (b) DC-DC-Converter

Figure 1.7: EMS applications examples

1.3.3 EMWorks2D

EMWorks2D is a low frequency simulation software that uses FEM to solve magnetic, electric and transient problems. Both planar and axisymmetric invariance geometries can be handled in 2D, which can shorten the design process. It enables as to test and improve our designs in record time. Having an intuitive workflow, EMWorks2D integrates seamlessly into the SOLIDWORKS environment, for a truly effortless and engaging simulation experience. EMWorks2D can be used to predict the performance of transformers, actuators, electric motors, generators, permanent magnet, inductors, magnetic gears, busbars, sensors, bushings, capacitors, insulators, wireless chargers and more...

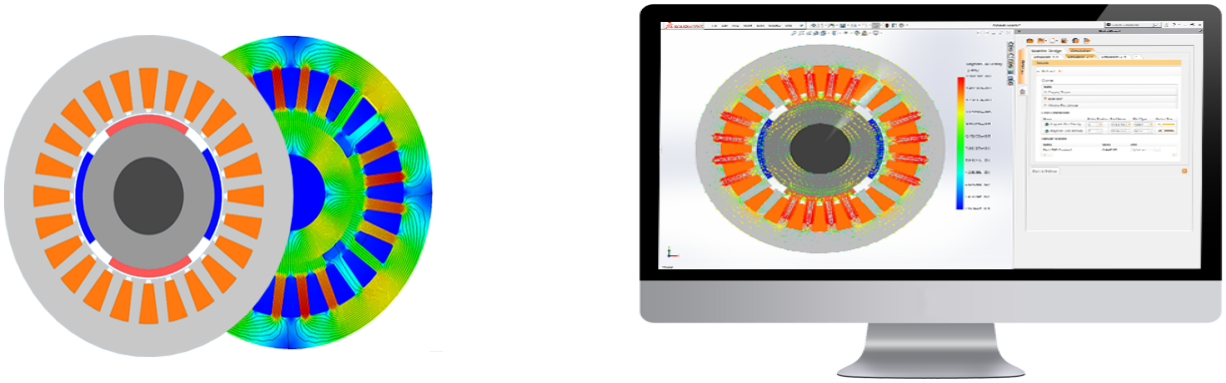


(a) Switched relectance motor (b) Transportation lines

Figure 1.8: EMWroks2D applications examples

1.3.4 MotorWizard

MotorWizard is a template-based motor design software which is completely integrated inside SOLIDWORKS. It allows SOLIDWORKS users to build and analyze different electric machine designs. It makes the study of the electric machines flexible and easy by giving access to a wide range of customizable dimensions and parameters that fully define the design of the electric machines. Equipped with integrated analytical and finite element-based solvers, the process of electric motor design becomes readily effortless, accurate and quick.



(a) Motor model

(b) MotorWizard-interface

Figure 1.9: EMWroks2D applications examples

1.4 Internship objectives

The main objective of this internship is to develop 3D and 2D simplified models of IPMSM used in electric cars and study their performance using finite element method simulation software. Software used in the simulation are EMS and EMWorks2D that are Embedded into SolidWorks

1.5 Conclusion

In this first chapter, the host company was introduced. It is a company specializing in electrical CAD-FEM based-on software. Among its large products, we had the opportunity to handle and master two software, namely EMS and EMWrks2D, which are both certified gold product SolidWorks. Then the project framework and its objectives are presented. Our end-of-studies project focuses on electric motors for the automotive industry based on finite element method electromagnetic simulation. The next chapter deals with 3D and 2D electromagnetic simulation under EMS and EMWorks2D software.

Chapter 2

Analysis and comparison of electric machines

2.1 Introduction

Electric motors are now more diverse and adaptable than ever before, each application requires a different type of motor, thus the choice of motor is extremely important. The motor must align with the purpose and overall performance goals of any given application. The next diagram regroups the most known electric motors. This chapter aims to introduce

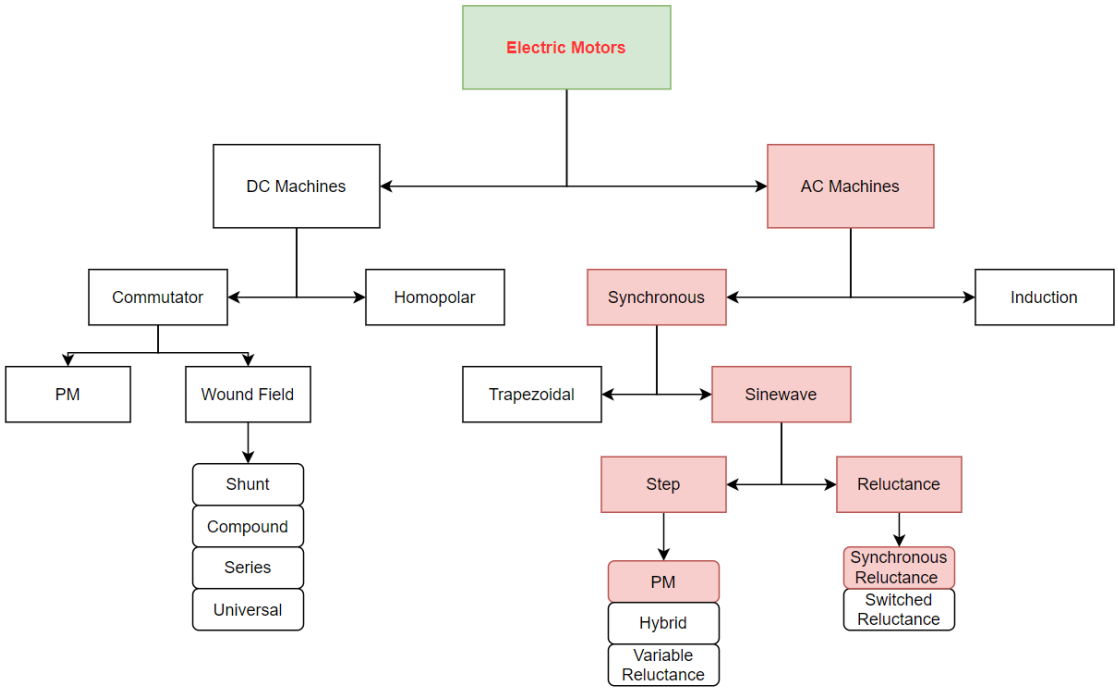


Figure 2.1: Diagram presenting different types of electric machines

the different types of AC electric motors, the most known types of magnetic materials and the necessary equations to understand how those machines works.For the purpose of our work, we will be just interested in machines that are highlighted in the diagram above, especially what are called brushless machines

2.2 Electric motors used in electric cars

2.2.1 Brushless motors

With the advancement in power electronics and IGBT technology, the use of Both brushless DC and brushless AC machines became more attractive.[6]
A brushless motor is, as its name suggests, a motor without brushes as in DC brushed motors or slip rings and mechanical commutators as in AC synchronous machines with wound rotor.

2.2.2 AC induction

The first machine that satisfies this basic definition is the AC induction motor with cage, in which current in the rotor windings is produced by electromagnetic induction. This type of motors employs a rotating magnetic field that rotates at a synchronous speed set by the supply frequency. However, the rotor rotates at a slightly slower speed because the process of electromagnetic induction requires relative motion, or slip, between the rotor conductors and the rotating field. so that those machines are called asynchronous, because the slip makes the rotor speed no longer proportional to the supply frequency.In addition, the induced rotor currents also give rise to eddy current losses which decrease the efficiency, in addition, The efficiency and power factor of induction motors falls off in small sizes . As a motor of given geometry is scaled down, if all dimensions are scaled at the same rate, the m.m.f. required to produce a given flux-density decreases in proportion to the linear dimension. The Tesla Roadster uses an induction motor as their main drive. [1].

2.2.3 Brushless permanent magnet synchronous motor

The brushless synchronous machine has permanent magnets instead of a field winding. Field control is sacrificed for the elimination of brushes, slip rings, and field copper losses. This motor is a 'classical' salient-pole synchronous a.c. motor with approximately sine distributed windings, and it can therefore run from a sinewave supply without electronic commutation. The magnets can be mounted on the rotor surface, or they can be internal to the rotor. The interior construction simplifies the assembly and relieves the problem of retaining the magnets against centrifugal force. It also permits the use of rectangular instead of arc-shaped magnets.[1]

2.2.4 Reluctance synchronous motor

The concept of the synchronous reluctance machine (Syn-RM) dates to the beginning of the previous century, however, only in recent years is gaining more attention. Despite advantages such as its low cost, simple, and rugged construction, cold rotor, a wide field-weakening range and the absence of expensive rare-earth metals, only recent progress in rotor high-anisotropy design has allowed the Syn-RM to achieve a better efficiency.[7]
A reluctance motor is a type of electric motor that induces non-permanent magnetic poles on the ferromagnetic rotor. The rotor does not have any windings. It generates torque through magnetic reluctance. We can also insert magnets inside the flux barriers and the motor become a reluctance permanent magnet assisted motor.

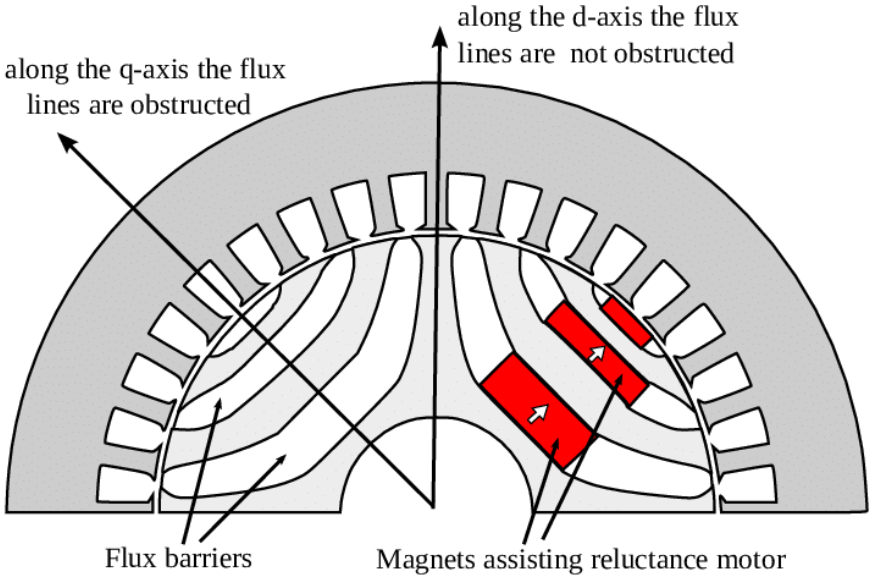


Figure 2.2: Synchronous Reluctance Machine

Synchronous reluctance motors are well suited for electric and Hybrid electric cars [8].

2.2.5 Interior permanent magnet synchronous motors

As it's mentioned above, the magnets in a synchronous machine can be either on the surface of the rotor lamination, surface mounted permanent magnet machines(**SPM**), or they can be buried inside the rotor, and thus we have what we call an internal synchronous permanent magnet motor (IPMSM).

Those machines are used in a wide range of industrial applications, especially for electric vehicles (EVs), attributing to the compact structure, high efficiency, high power/torque density and excellent dynamic performances [10].

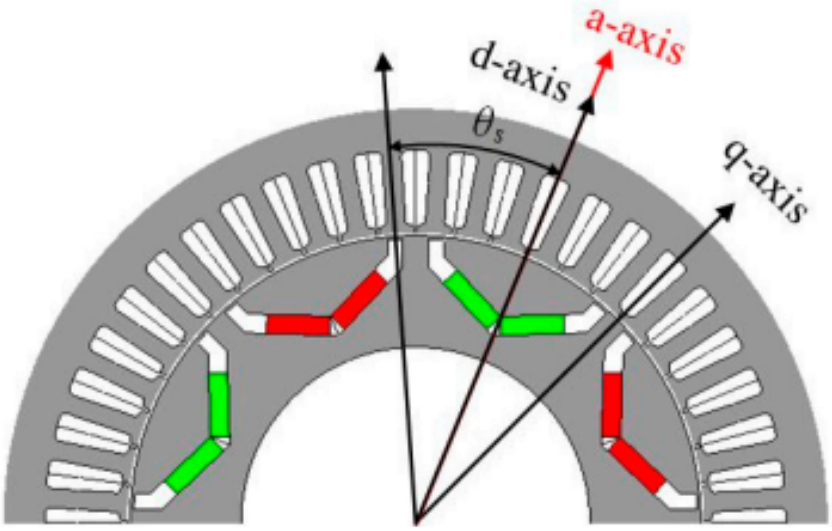


Figure 2.3: Interior Permanent Magnet Motor Machine [9]

2.3 Magnetic materials

2.3.1 Permanent magnets materials and caracteristics

The evolution of permanent magnets' development is illustrated in figure 2.4 [11]. the latest

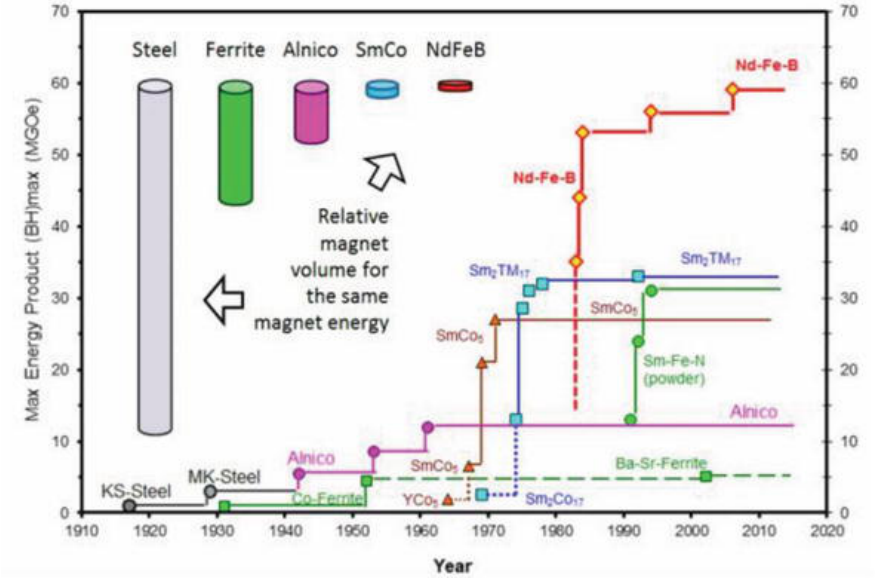


Figure 2.4: Permanent magnets evolution

addition being neodymium-iron-boron which has been pioneered by Sumitomo as 'Neomax',

General Motors as 'Magnequench', Crucible ('Crumax'), and IG Technologies ('NelGT'). At room temperature NdFeB has the highest energy product of all commercially available magnets(there are multiple grades of neodymium such as N30-N52,N30M-N50M ...) in which they are graded according to maximum energy products. The high remanence and coercivity permit marked reductions in motor framesize for the same output compared with motors using ferrite (ceramic) magnets. However, ceramic magnets are considerably cheaper. [1]. Magnets proprieties are summarized in the following table :

Proprety	Units	Alnico 5-7	Ceramic	Sm_2Co_1	NdFeB
B_r	T	1.35	0.405	1.06	1.12
$\mu_0 H_c$	T	0.0074	0.37	0.94	1.06
$(BH)_{max}$	MGO_e	7.5	3.84	26.0	30.0
μ_{rec}		1.9	1.1	1.03	1.1
Specific gravity		7.31	4.8	8.2	7.4
Resistivity	$\mu\Omega cm$	47	$> 10^4$	86	150
Thermal Expansion	$10^-6/Celcius$	11.3	13	9	3.4
B_rTemperature	%/Celsius	-0.02	-0.2	-0.025	-0.1
Saturation H	kOe	3.5	14	> 40	> 30

Table 2.1: Brief summary of magnet proprieties [1]

2.3.2 Hysteresis loop

The starting point to understand magnet characteristics is understanding the hysteresis loop [1]. The x-axis measures the magnetizing force or the 'field intensity' H in the material. The y-axis is the magnetic flux-density B in the material.
An demagnetized sample has B=0 and H=0 and therefore starts out at the origin If it is subjected to a magnetic field, then B and H in the magnet will follow the curve OA as the external ampere-turns are increased. If the external ampere-turns are switched off, the magnet relaxes along AB. Its operating point (H, B) will depend on the shape of the magnet and the permeance of the surrounding 'magnetic circuit'. If the magnet is surrounded by a highly permeable magnetic circuit, that is, if it is 'keepered', then its poles are effectively shorted together so that H=0 and the flux-density is then the value at point B, the **remanence** B_r. The remanence is the maximum flux-density that can be retained by the magnet at a specified temperature after being magnetized to saturation. External ampere-turns applied in the opposite direction cause the magnet's operating point to follow the curve from B through the second quadrant to C, and again if they are switched off at C the magnet relaxes along CD. It is now magnetized in the opposite direction, and the maximum flux-density it can retain when 'keepered' is B_r. To bring the flux-density to zero from the original positive

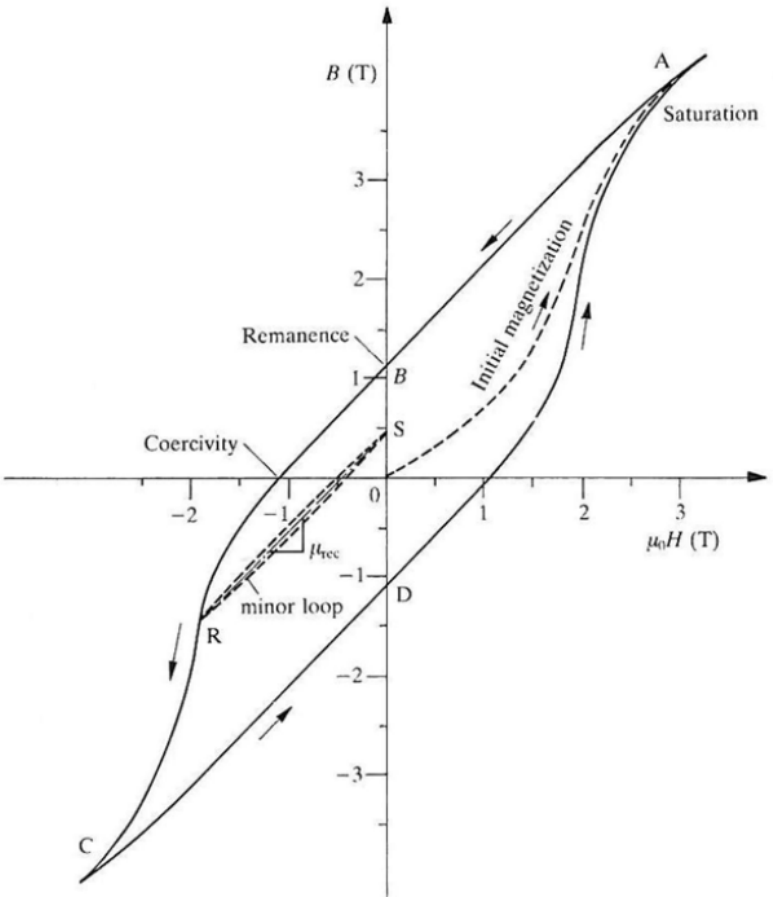


Figure 2.5: Permanent magnet hysterisis Loop

remanence, the external ampere-turns must provide within the magnet a negative magnetizing force $-H_c$, called the **coercivity**. Likewise, to return the flux-density to zero from the negative remanence point D, the field $+H_c$ must be applied.

2.4 Mathematical modelling of IPMSM

The steady state performance of interior PMSMs can be modelled by circuit equations with stator PM flux linkage and DQ axis inductances that are non-linear functions of the machine flux level.

2.4.1 IPMSM equivalent circuit and phasor diagram

The equations in terms of phase variables can be written as :

$$V_a = R_s I_a + \frac{d\lambda_a}{dt} \tag{2.1}$$

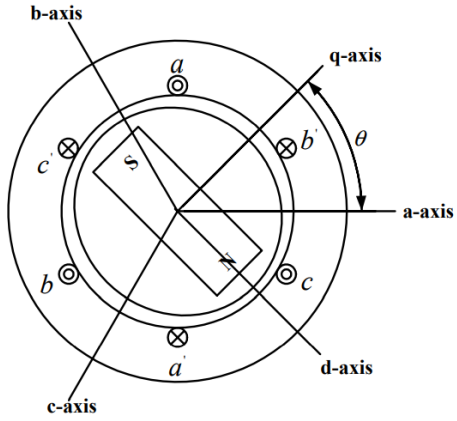


Figure 2.6: A Bipolar salient pole PMSM

$$V_b = R_s I_b + \frac{d\lambda_b}{dt} \tag{2.2}$$

$$V_c = R_s I_c + \frac{d\lambda_c}{dt} \tag{2.3}$$

Where:
V_a, V_b, V_c are 3 phase voltages
I_a, I_b, I_c are the 3 phase currents
R_s phase resistance
$\lambda_a, \lambda_b, \lambda_c$ are the 3 phase flux linkage
The 3 phase flux linkage equations are :[12]

$$\lambda_a = L_{aa} I_a + L_{ab} I_b + L_{ac} I_c + \lambda_{ma} \tag{2.4}$$

$$\lambda_b = L_{ba} I_a + L_{bb} I_b + L_{bc} I_c + \lambda_{mb} \tag{2.5}$$

$$\lambda_c = L_{ca} I_a + L_{cb} I_b + L_{cc} I_c + \lambda_{mc} \tag{2.6}$$

Where:
$\lambda_{ma}, \lambda_{mb}, \lambda_{mc}$correspond to the phase flux linkage provided by the permanent magnets.
In these equations, inductances are functions of the angle θ. Since stator self inductances are maximum when the rotor q-axis is aligned with the phase axis, while mutual inductances are maximum when the rotor q-axis is in the midway between two phases [13],
Flux linkage due to the permanent magnets are:

$$\lambda_{ma} = \lambda_m \cos\theta \tag{2.7}$$

$$\lambda_{mb} = \lambda_m \cos\left(\theta - \frac{2\pi}{3}\right) \tag{2.8}$$

$$\lambda_{mc} = \lambda_m \cos\left(\theta + \frac{2\pi}{3}\right) \tag{2.9}$$

For this model, input power P_{in} can be represented as:

$$P_{in} = \vec{V_a}\vec{I_a} + \vec{V_b}\vec{I_b} + \vec{V_c}\vec{I_c} \tag{2.10}$$

By using Park and Clarke transformation we can transform the (abc) reference frame quantities to the (DQZ) frame to remove the effect of time on the inductances, and thus they become only dependent on θ.

$$u_{DQZ} = M_{CP}.u_{abc} \tag{2.11}$$

Where u denotes either the voltage, the flux linkage, or the inductances, and M_{CP} is the Clarke-Park matrix.

$$M_{CP} = \sqrt{\frac{2}{3}} \begin{bmatrix} \cos\theta & -\sin\theta & \frac{\sqrt{2}}{2} \\ \cos(\theta - \frac{2\pi}{3}) & -\sin(\theta - \frac{2\pi}{3}) & \frac{\sqrt{2}}{2} \\ \cos(\theta + \frac{2\pi}{3}) & -\sin(\theta + \frac{2\pi}{3}) & \frac{\sqrt{2}}{2} \end{bmatrix} \tag{2.12}$$

By using this transformation, the phase voltage in the DQ frame are :

$$v_d = Ri_d + \frac{d\lambda_d}{dt} - \omega_s\lambda_q \tag{2.13}$$

$$v_q = Ri_q + \frac{d\lambda_q}{dt} + \omega_s\lambda_d \tag{2.14}$$

where :

$$\lambda_q = \lambda_{aq} = L_q I_q \tag{2.15}$$

$$\lambda_d = \lambda_{ad} + \lambda_m = L_d i_d + \lambda_m \tag{2.16}$$

$\lambda_m$is the permanent magnet flux linkage



(a) equivalent q axis circuit of PMSM

(b) equivalent d axis circuit of PMSM

Figure 2.7: equivalent circuit of PMSM in DQ reference frame

According to the equivalent circuit in figure 2.7 the phasor diagram can be constructed easily (2.8).

We can extract the input power equation :

$$P_{in} = \begin{bmatrix} V_a & V_b & V_c \end{bmatrix} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} = \frac{3}{2} \begin{bmatrix} V_q & V_d \end{bmatrix} \begin{bmatrix} i_q \\ i_d \end{bmatrix} = \frac{3}{2}(V_q i_q + V_d i_d) \tag{2.17}$$

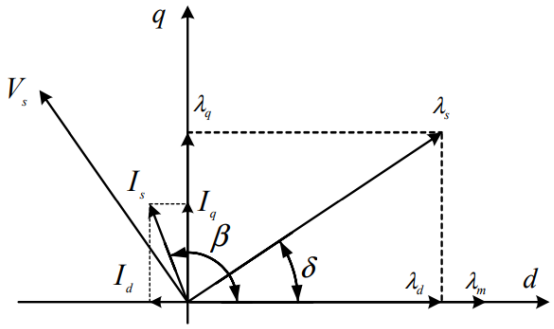


Figure 2.8: Phasor diagram [13]

$$P_{in} = \frac{3}{2}((Ri_q + \frac{d\lambda_q}{dt} + \omega_s\lambda_d)i_q + (Ri_d + \frac{d\lambda_d}{dt} - \omega_s\lambda_q)i_d) \tag{2.18}$$

By neglecting the phase resistance $R = 0$ and supposing that the inductance L_q, L_d are time invariant since we are determining a (steady state) equation, we obtain :

$$P_0 = \frac{3}{2}(-\omega_s\lambda_q i_d + \omega_s\lambda_d i_q) \tag{2.19}$$

inserting equations 2.15 and 2.16 we obtain:

$$P_0 = \frac{3}{2}(-\omega_s L_q i_q i_d + \omega_s(L_d i_d + \lambda_m)i_q) \tag{2.20}$$

$$P_0 = \frac{3}{2}(-\omega_s L_q i_q i_d + \omega_s L_d i_d i_q + \omega_s\lambda_m i_q) \tag{2.21}$$

$$P_0 = \frac{3}{2}(-\omega_s(L_q - L_d)i_q i_d + \omega_s\lambda_m i_q) \tag{2.22}$$

$$\frac{P_0}{\omega_s} = \frac{3}{2}(-(L_q - L_d)iqid + \lambda_m i_q) = \frac{3}{2}((L_d - L_q)i_d i_q + \lambda_m i_q) \tag{2.23}$$

Then we can determine the Torque since it is power divided by mechanical speed:

$$T_e = \frac{P_0}{\omega_m} \tag{2.24}$$

where ω_m is the mechanical speed
we know that the electrical and mechanical relations are related with number of pole pair in electric motors :

$$\theta_s = P\theta_m \tag{2.25}$$

The electric position is defined such that the movement of the rotor by 360electric degree puts the rotor in the same magnetic orientation .for example, if the motor is 2 poles motor than we must rotate the rotor 360 mechanical to put the rotor in the same magnetic orientation,

if the motor is a 4 pole machine than we have to rotate the rotor 180 mechanical to put the rotor in the same magnetic orientation. Thus, $360E = 180mechanical$. This explains the above equation. From that equation, we can determine the relationship between the mechanical speed and the electrical speed by taking its time derivative.[14]

$$\omega_s = p\omega_m \tag{2.26}$$

finely, the electromagnetic torque equation of a P pair pole machine is :

$$T_e = \frac{P_0}{\omega_m} = \frac{P_0}{\frac{\omega_s}{P}} = P\frac{P_0}{\omega_s} = \frac{3P}{2}[\lambda_m I_q + (L_d - L_q) i_d i_q] \tag{2.27}$$

From this equation, we can see that the Torque is constructed with two different terms:
$\lambda_m I_q$ is the magnet torque and the $(L_d - L_q) I_d I_q$ term is the reluctance torque due to the difference of the direct and quadrature inductances, and hence it proves that the IPMSM is a hybrid combination of the reluctance synchronous machine and the conventional permanent magnet synchronous machine.

2.4.2 Finite element method for electromagnetism

The finite element method was proposed in the 1940s, but it was firstly applied almost ten years later in aeronautical design and in structural analysis. After a couple of years, it was largely adopted in almost all physical and mathematical problems. The finite element method for electromagnetism is essentially based on the determination of the distribution of the electric and magnetic fields in the structures under study, based on the solution of Maxwell's equations. because the complexity of electric machines structure and the non-linearity of materials analytical solution are barely achieved, thus, in most cases, only a numerical solution is possible.The finite element method is a numerical technique that is suitable for this purpose. It allows a field solution to be obtained, even with time-variable fields and with materials that are non homogeneous, anisotropic, or non-linear. Using the finite element method, the whole analysis domain is divided into elementary subdomains, which are called finite elements, and the field equations are applied to each of them.[15]

Fundamental equations summary

All the electromagnetism theory is governed by Maxwell's equations :

$$\boldsymbol{\nabla} \times \vec{H}(P,t) = \vec{J} + \frac{\partial \vec{D}(P,t)}{\partial t} \tag{2.28}$$

$$\boldsymbol{\nabla} \times \vec{E}(P,t) = -\frac{\partial \vec{B}(P,t)}{\partial t} \tag{2.29}$$

$$\boldsymbol{\nabla} \cdot \vec{B}(P,t) = 0 \tag{2.30}$$

$$\boldsymbol{\nabla} \cdot \vec{D}(P,t) = \rho(P,t) \tag{2.31}$$

where :
$\vec{H}(P,t)$ is the magnetic field strength
$\vec{D}(P,t)$ is the electric displacement
$\vec{E}(P,t)$ is the electric field strength
$\vec{B}(P,t)$ is the magnetic flux density
There are also the constitutive relationships and the continuity equation :

$$\vec{B}(P,t) = \mu(P)\vec{H}(P,t) \tag{2.32}$$

$$\vec{D}(P,t) = \epsilon(P)\vec{E}(P,t) \tag{2.33}$$

$$\vec{J}(P,t) = \sigma(P)\vec{E}(P,t) \tag{2.34}$$

and the continuity equation :

$$\boldsymbol{\nabla} \cdot \vec{J}(P,t) = -\frac{\partial \rho(P,t)}{\partial t} \tag{2.35}$$

$$\vec{B}(P,t) = \boldsymbol{\nabla} \times A(P,t) \tag{2.36}$$

$A(P,t)$ is the magnetic vector potential

Laplace, Poisson equations

From equation 2.36 and 2.30 we can have the Laplace equation :

$$\boldsymbol{\nabla} \times \left(\frac{1}{\mu} \boldsymbol{\nabla} \times A \right) = J \tag{2.37}$$

If the material is homogenous and the permeability μ is constant then 2.37 can be written :

$$\nabla^2 \vec{A} = -\mu \vec{J} \tag{2.38}$$

which is Poisson equation

Field problem with boundary conditions

Generally, a vector field problem is described by a differential equation, defined in a domain D:[15]

$$L\phi(P,t) = f(P,t) \tag{2.39}$$

Where :
L is a differential operator
ϕ is the unknown function we are determining
f is the forcing function
L might be any differential operator

In the electromagnetic problems' equation 2.39 is given by Laplace or Poisson Equation, for example in an electrostatic problem ϕ is the scalar electric potential V and its distribution is described the Poisson equation and the forcing function f is the charge density ρ . $f = \rho$ and equation 2.39 can be written as :

$$-\boldsymbol{\nabla} \cdot \left(\epsilon \vec{\nabla} V\right) = \rho \tag{2.40}$$

in this equation L is :

$$L = \boldsymbol{\nabla} \cdot \left(\epsilon \vec{\nabla}\right) \tag{2.41}$$

for resolving this differential equation, boundary conditions must be applied to assert the unicity of the solution
there is the Dirichlet boundary condition, it's the assignment of a value of ϕ on a boundary Γ

$$\phi = cte \tag{2.42}$$

and there is the Neumann condition which is when a given value of the derivative of ϕ normal to Γ is assigned.

$$\frac{\partial \phi}{\partial n} = cte \tag{2.43}$$

Methods for resolving field problems

There are three main methods for resolving the equation 2.39, with a given boundary condition, that are : **Galerkin's method**, also called the **Residual method**,**Rayleigh-Ritz's method**, also called **Variational, method** and, **The finite element method**.
All those methods aim to define a new function ϕ^* that approximates ϕ as closely as possible, generally ϕ^* is expressed as a linear combination of basic functions [15].

$$\phi^*(P,t) = \sum_{j=1}^{N} \Phi_j \nu_j(P,t) \tag{2.44}$$

Where ν_j are interpolating functions and Φ_j are unknown coefficients that will be determined during the computation process.
the residual method and the variational method, take into account the whole analysis domain. The functions ν_j are defined on the whole domain. Conversely, in the finite element method, the whole domain is divided in subdomains.
The Galerkin's method or the Rayleigh-Ritz's method is a prerequisite for the finite element method ,we will be just describing the Galerkin's method

Galerkin's method

before describing the method, the L operator must verify the inner scalar product proprieties.

$$< \phi, \varphi >= \int_{\tau} L\phi\tilde{\varphi} d\tau = \begin{cases} > 0 & \phi \neq 0 \\ = 0 & \phi = 0 \end{cases} \tag{2.45}$$

$$< L\phi, \varphi >=< \varphi, L\phi > \tag{2.46}$$

where :
$\tilde{\varphi}$ is the complex conjugate of φ and τ is the volume .

The resolution of equation 2.39 using the Galerkin's method consists of reducing its residual.

$$r = L\phi^* - f \tag{2.47}$$

One way to do that is by using some weighted functions ω_i than the residual method forces the integral of the residuals weighted by ω_i to be 0.

$$R_i = \int_{\tau} \omega_i (L\phi^* - f) d\tau = 0 \tag{2.48}$$

in the Galerkin's method the weights ω_i are equal to the interpolating functions ν_i

$$\omega_i = \nu_i \tag{2.49}$$

The finite element method

As described above, the finite element method consist of subdividing the hole domain D into N_m subdomains D_m , generally called mesh elements,

in a one dimensional problem the domain is a curve and the subdomain is a line



Figure 2.9: line element

In two-dimensional problems, the domain is a surface and each subdomain is a polygon, usually a triangle or a rectangle.

And in three-dimensional problems, the domain is a volume and each subdomain is a tetrahedron, a triangular prism, or a rectangular solid.

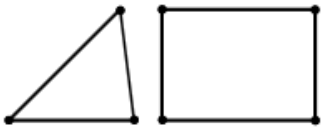
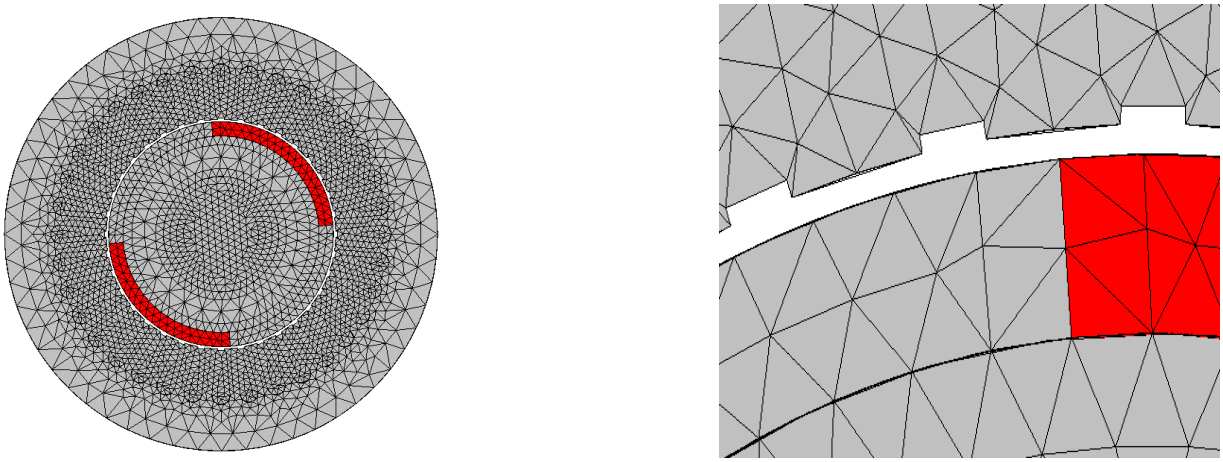


Figure 2.10: mesh element in 2D



(a) 2D mesh from EMworks 2D

(b) Mesh Zoom

Figure 2.11: 2D meshed Domain with EM2D

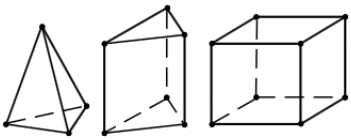


Figure 2.12: mesh element in 3D

Once the domain is meshed, either the residual or the variational method is used to calculate solution of the field problem.
The calculation techniques are more complex to present, they are fully integrated into FEA software. for further details about how computations are done, A numerical example is presented in Appendix A

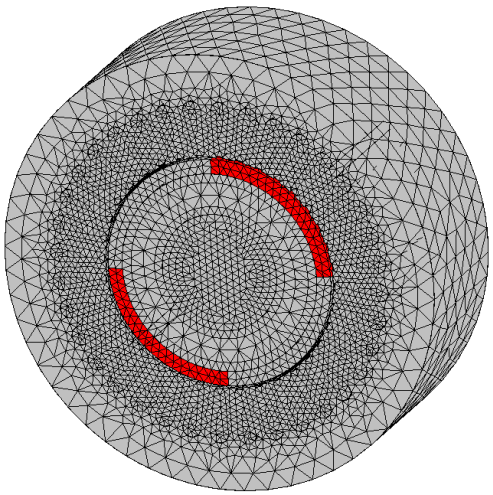


Figure 2.13: a 3D meshed domain with EMS

2.5 Torque calculation using virtual work method and Maxwell stress tensor

2.5.1 Maxwell stress tensor method

The Maxwell stress tensor is a symmetric second-order tensor used in classical electromagnetism to represent the interaction between electromagnetic forces and mechanical momentum. for determining the Maxwell stress tensor, we start with the Lorentz force [16]

$$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B}) \tag{2.50}$$

the force per unite volume is :

$$\vec{f} = \rho\vec{E} + \vec{J} \times \vec{B} \tag{2.51}$$

and by using Ampere law and Gasses law, equation 2.51 becomes

$$\vec{f} = \epsilon_0(\boldsymbol{\nabla} \cdot \vec{E})\vec{E} + \frac{1}{\mu_0}(\boldsymbol{\nabla} \times \vec{B}) \times \vec{B} - \epsilon_0 \frac{\partial \vec{E}}{\partial t} \times \vec{B} \tag{2.52}$$

The time derivative can be rewritten to something that can be interpreted physically, namely the Pointing vector.

$$\frac{\partial}{\partial t}\vec{E} \times \vec{B} = \frac{\partial \vec{E}}{\partial t} \times \vec{B} + \frac{\partial \vec{B}}{\partial t} \times \vec{E} = \frac{\partial \vec{E}}{\partial t} \times \vec{B} - \vec{E} \times (\boldsymbol{\nabla} \times \vec{B}) \tag{2.53}$$

Hence, we can rewrite 2.52 as follows

$$\vec{f} = \epsilon_0(\boldsymbol{\nabla} \cdot \vec{E})\vec{E} + \frac{1}{\mu_0}(\boldsymbol{\nabla} \times \vec{B}) \times \vec{B} - \epsilon_0 \frac{\partial}{\partial t}(\vec{E} \times \vec{B}) - \epsilon_0 \vec{E} \times (\boldsymbol{\nabla} \times \vec{E}) \tag{2.54}$$

By regrouping the terms of $\vec{E}$ and $\vec{B}$ and using Gasses law for magnetism to insert the $(\boldsymbol{\nabla} \cdot \vec{B})\vec{B}$ we obtain :

$$\vec{f} = \epsilon_0[(\boldsymbol{\nabla} \cdot \vec{E})\vec{E} - \vec{E} \times (\boldsymbol{\nabla} \times \vec{E})] + \frac{1}{\mu_0}[(\boldsymbol{\nabla} \cdot \vec{B})\vec{B} - \vec{B} \times (\boldsymbol{\nabla} \times \vec{B}) - \epsilon_0 \frac{\partial \vec{E}}{\partial t}] \tag{2.55}$$

and by using :

$$\frac{1}{2}(\boldsymbol{\nabla} \cdot \vec{A}).\vec{A} = \vec{A} \times (\boldsymbol{\nabla} \times \vec{A}) + (\vec{A}.\nabla)\vec{A} \tag{2.56}$$

$\vec{f}$ becomes :

$$\vec{f} = \epsilon_0[(\boldsymbol{\nabla}\cdot\vec{E})\vec{E}+(\vec{E}.\nabla)\vec{E}]+\frac{1}{\mu_0}[(\boldsymbol{\nabla}\cdot\vec{B})\vec{B}+(\vec{B}\nabla)\vec{B}]-\frac{1}{2}\nabla(\epsilon_0 E^2+\frac{1}{\mu_0}B^2)-\epsilon_0\frac{\partial}{\partial t}(\vec{E}\times\vec{B}) \tag{2.57}$$

and now by introducing the Maxwell stress tensor

$$\sigma_{ij}^M = \epsilon_0(E_i E_j - \frac{1}{2}\delta_{ij}E^2) + \frac{1}{\mu_0}(B_i B_j - \frac{1}{2}\delta_{ij}B^2) \tag{2.58}$$

and using the Pointing vector $\vec{S} = \frac{1}{\mu_0}\vec{E} \times \vec{B}$ we obtain :

$$\boldsymbol{\nabla} \cdot \vec{\sigma^M} = f + \epsilon_0\mu_0\frac{\partial \vec{S}}{\partial t} \tag{2.59}$$

This method generally yields good results, but it is very sensitive to local errors which make it unstable, especially in very small air gaps. For this reason, EMS does not use it for force calculation(source:EMWorks user guide)

2.5.2 Virtual work method

This method consists of finding the change in the total magnetic energy when the object is displaced a distance Δs in the direction of the force component:

$$F_s = \frac{W_2 - W_1}{\Delta s} \tag{2.60}$$

where:

$$W = \frac{1}{2}\int \vec{B}.\vec{H}dv \tag{2.61}$$

is the magnetic energy . The torque is obtained by rotating the component a $\delta\phi$ around x-, y-, or z-axis:

$$T_{axis} = \frac{W_2 - W_1}{\delta\phi} \tag{2.62}$$

source EMWroks user guide.

2.6 Conclusion

Electric machines, especially electric motors, are widely used, and they reviled their importance in a lot of applications, with the need to advance toward electrification we must engineer electric motors with high capabilities, Numerical methods revile their importance in analyzing and optimizing electric machines as we will see in chapter 3, thus understanding those methods is a need for any electrical engineer.

Chapter 3

Numerical Study and results

3.1 Introduction

In this chapter, we will be regrouping the simulation results of the Tesla Model 3 and Toyota Prius IPMSM's motors, a 3D versus 2D simulation is illustrated to demonstrate the power of EMworks2D when using symmetrical models.We will also see the effect of rotor geometry modification and magnet type on material saturation Torque.

3.2 2D vs 3D finite element analysis with EMS and EM2D

3.2.1 Design of a 3D simplified model of PMSM

The first step to take in every simulation process is to prepare the model. a modified design from [17] of a 36 slot 4 pole Permanent magnet synchronous motor (PMSM) is used. the motor dimensions are summarized in table 3.1 For further details refer to Appendix C

Propriety	value
Number of stator slots	36
Stator lamination outer diameter	117mm
Stator lamination inner diameter	79.5
Rotor outer diameter	78mm
Rotor inner diameter	40mm
Lamination stuck length	50mm

Table 3.1: PMSM dimensions

Propriety	value
Total depth of each slot	19.25
Maximum slot width	$a_2 = 6.35mm$
slot width	$a_1 = 3.52mm$
Slot opening	$e = 2.25mm$
slot depth	$b = 14.825mm$
Slot depth	$c = 0.75$
Slot depth	$d = 1.5$

Table 3.2: slot dimensions of PMSM

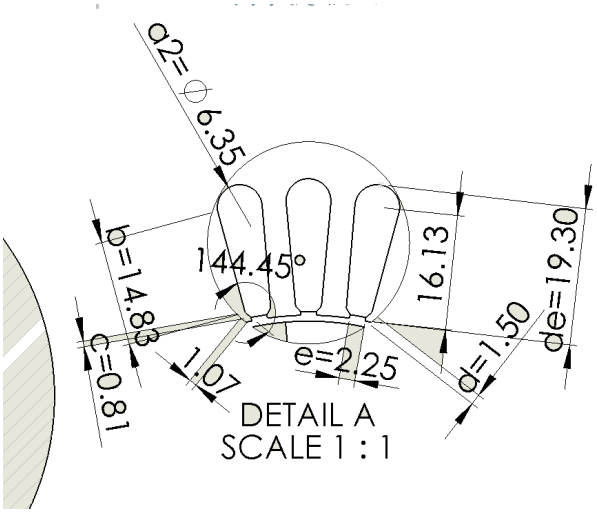


Figure 3.1: PMSM slot dimensions (see Appendix C)

Core and magnet material

The core and magnets materials of both motors are summarized in table 3.3.

	Material
Stator	Stainless Steel 1010
Rotor	Stainless Steel 1010
magnet	Samarium-cobalt Magnet S2818

Table 3.3: Materials of PMSM

3D geometry design

The 3D model was designed using SolidWorks, and the 2D model was obtained using a powerful tool from Emworks2D, which is the 2D simplification.

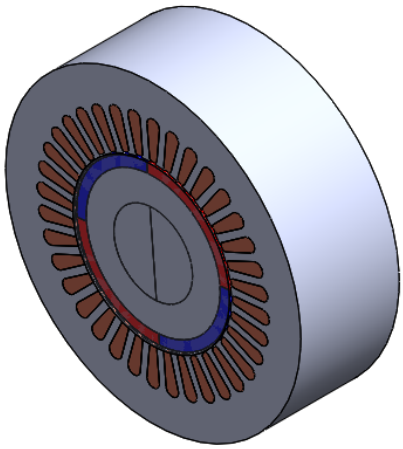
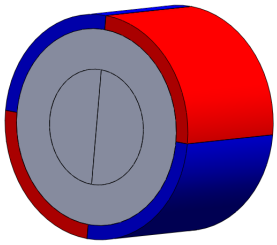
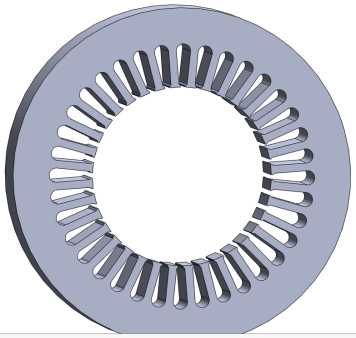


Figure 3.2: PMSM assembly



(a) Rotor of PMSM



(b) Stator of PMSM

Figure 3.3: PMSM rotor and stator

once the parts are ready we prepare the assembly, in this section a 3D transient magnetic analysis is used coupled with motion. the parts are illustrated in the next figures :

3.2.2 No load results comparison

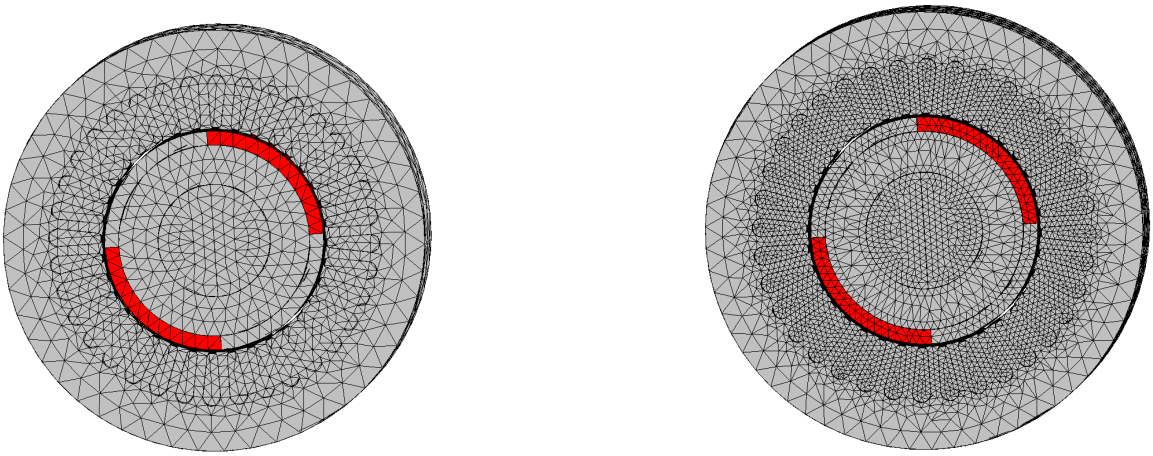
Mesh

The finer the mesh, the more accurate results are.
Using finer mesh in a model will require more computing resources, but 3D models almost

always requires more resource even if the mesh is coercer compared with using a finer mesh in 2D.

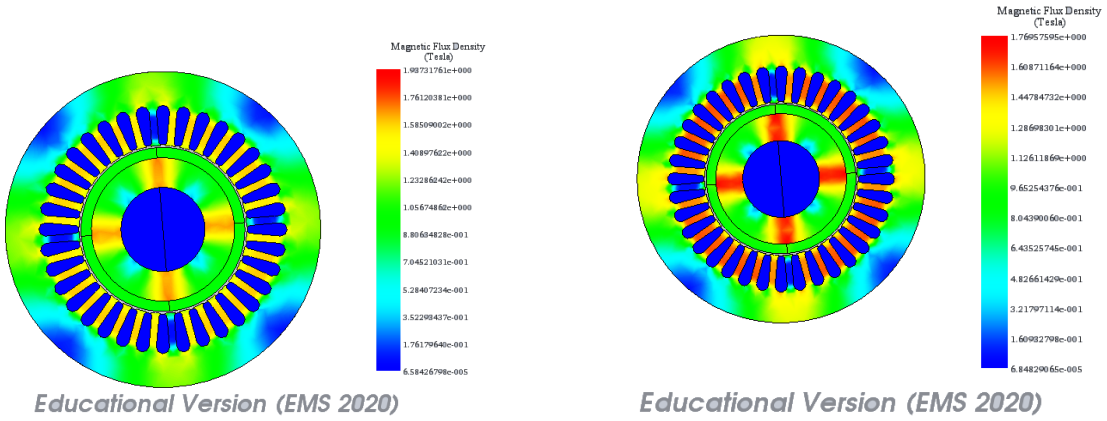
The mesh size was chosen such that the computer can handle the simulation and the results are fairly acceptable. we will be using 2 different mesh elements to see how results change.

figure 3.4 represents 2 different meshes used during the simulation.



(a) Coerce mesh (b) finer mesh

Figure 3.4: 2 Different mesh domains with EMS of PMSM



(a) flux density of PMSM with coerce mesh (b) flux density of PMSM with finer mesh

Figure 3.5: flux density using different mesh of a PMSM

We could see from figures 3.5 that the mesh size affects deeply the simulation results, the values are largely dispersed, by using finer we could see that the rotor core was saturated by the magnets in figure 3.5b as opposite to figure 3.5a hence to obtain accurate results the

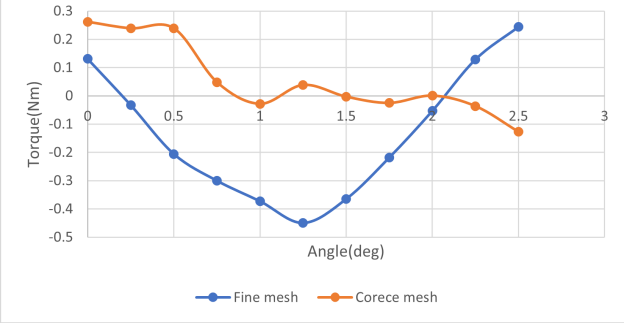


Figure 3.6: Cogging Torque comparison with different mesh size

mesh must be as fine as possible which in turns need more computation resources as we said above.

2D vs 3D no load torque comparison

Figure 3.7 represents the No load torque (which is a cogging torque) comparison between the 3D and 2D model.

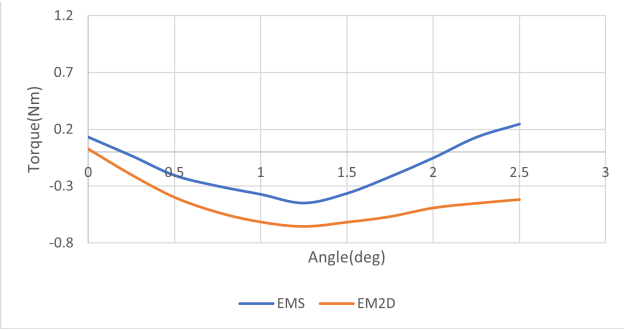
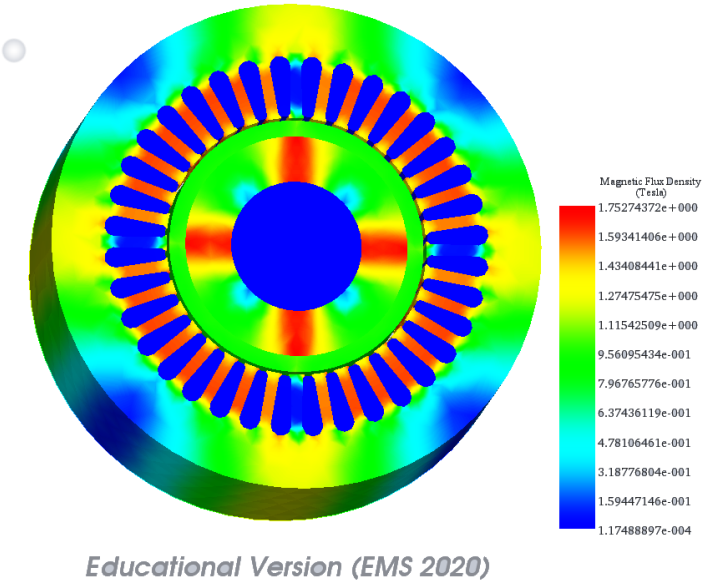


Figure 3.7: Cogging torque

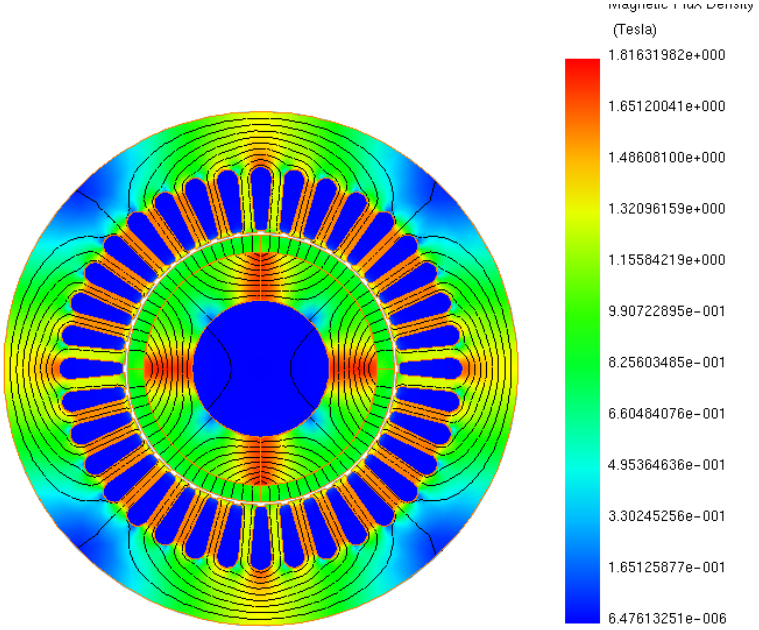
We can see from this figure that the 3D model gave more accurate results and contains more details, nevertheless, the 2D model gave a fairly acceptable results.

Figure 3.40 presents the flux density distribution of both models, we can see that the material of the rotor and stator teeth are a bit saturated by the magnets in both the 3D and 2D model, the cogging torque wave form are approximately identical.

this simulation was set on a Terra mobile, dotted with an i73632QM processor and 8 GB RAM DDR3, with finer mesh EMS uses the entire memory and swap, in the other hand EMWorks2D used approximately the half of the computation resources and simulation time,this demonstrates that when having a symmetrical model, EMWorks2D is very powerful, and it gives accurate results. In the next section, we will be using EMWorks2D.



(a) 3D flux density distribution of PMSM



(b) 2D flux density distribution of PMSM

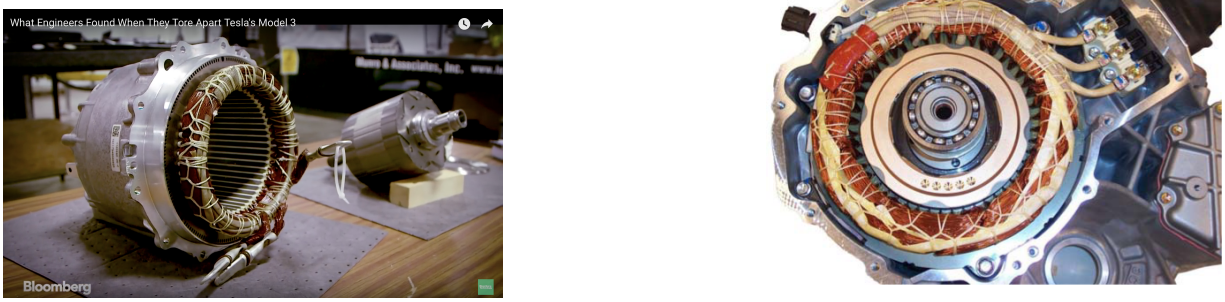
Figure 3.8: Flux density

3.3 Toyota Prius 2004 and The Tesla Model 3 2018

The 2004 Prius is a new-generation hybrid automobile that was introduced into the market in September 2003 by the Toyota Motor Corporation. As a hybrid vehicle, the 2004 Prius uses both a gasoline-powered internal combustion engine capable of delivering a peak-power output of 57 kW and a battery-powered electric motor capable of delivering a peak-power output of 50 kW as motive power sources.
In the other hand, the Tesla Model 3 is a full electric car, and it was introduced in 2017, limited production of the Model 3 began in mid-2017, the single drive system delivers 190-211kW and 420-450Nm at its peak.

Motor	Tesla Model 3	Toyota Prius
Maximum output	190-211kW	50kW
Maximum torque	420-450Nm	400Nm
Maximum speed	12000RPM	10000 RPM
Nominal speed	6000RPM	1540RPM
Minimum bus voltage	213V	200V
Supply current	953ARMS	230ARMS

Table 3.4: IPMSM motor performance



(a) Tesla Model 3 2018 IPMSM

(b) Toyota Prius 2004 IPMSM

Figure 3.9: Toyota and Tesla real motors

3.3.1 Materials

The magnetic materials characteristics have been presented in Appendix C Core materials and magnets are characterized with their hysteresis loop, demagnetization curves of used magnets are also presented.

Motor	Tesla Model 3	Toyota Prius
Stator core material	M-27	M-19
Rotor core material	M-27	M-19
Magnets	N52	N42
coil	copper	copper

Table 3.5: IPMSM materials

Motor	Tesla Model 3	Toyota Prius
Stator iron	15.991Kg	17.731Kg
Stator winding	3.294Kg	5.678Kg
Rotor iron	10.904Kg	5.525Kg
Magnets	1.658Kg	1.174kg
Total mass	31.875Kg	29.808Kg

Table 3.6: IPMSM material weights

The materials weights have been determined using SolidWorks material propriety. The Tesla model which is an approximation of the real motor weights more than the Toyota motor, since The N52 is a higher grade in terms of high performance than it will be more expensive. We can for instance compare those motors based on their no load results and on load results.

3.3.2 Geometry specification

The IPMSM of the Toyota Prius 2004 has 48 stator slots, 8-V shaped poles buried inside the rotor lamination core, Both rotor and stator cores are made of laminated steel M-19 (stacking factor is 0.98). This motor has 3 phases, single layer distributed windings, 9 turns per coil and 1 parallel paths.
Whereas The IPMSM of The Tesla model 3 (2018) has 54 slots and 6 poles buried inside the rotor lamination core. Both rotor and stator cores are made of laminated steel M-27 (stacking factor is 0.95). This motor has 3 phases, double layered distributed windings, 2 turns per coil and 3 parallel paths.

it's clear that both motors are different geometrically and in terms of power needs, the stator and rotor core dimensions are summarized in table 3.8 the stack length and the diameter of the rotor are important because they affect the overall torque produced on the rotor yoke. By using fewer poles, we can minimize the diameter of the rotor, but the length must be increased to obtain the needed torque. Dimensioning of electric machines is generally

analytical. Some dimensioning techniques are presented in appendix C.

Stator	Tesla Model 3	Toyota Prius
Number of stator slots	54	48
Stator outer diameter	225mm	229.6
Stator inner diameter	151.3mm	160.5mm
Stator lamination stack length	134mm	83.6
Axial overhang	40mm	–
Lamination stacking factor	0.95	0.98
Number of phases	3	3
Number of parallel paths	3	1
number of turns percoil	2	9
Number of pole pair	3	4
Rotor	**Tesla Model 3**	**Toyota Prius**
Rotor outer diameter	149.9mm	160.5mm
Rotor inner diameter	70mm	111mm
Rotor lamination length	134mm	86.3mm

Table 3.7: Tesla and Toyota IPMSM Dimensions

for further details, refer to Appendix C.

3.3.3 CAD modeling

once we have the dimensions, we prepare the 3D models. The design was carried out using SolidWorks. the next figures show some steps of designing procedure of the Tesla motor. The models have been, means that we did not use the axial overhang since the rotor of the Tesla does not have the same stack length as the stator, also the coils have been simplified for the purpose of using symmetrical model.

The stator and rotor 3D models are presented below.

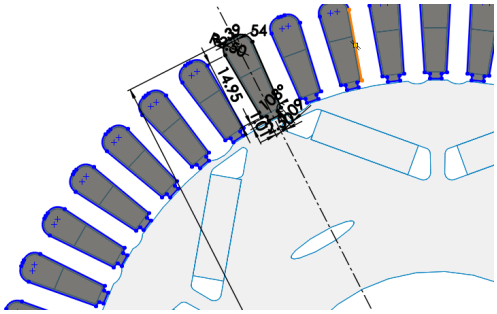


Figure 3.10: Tesla IPMSM slot dimensioning and design

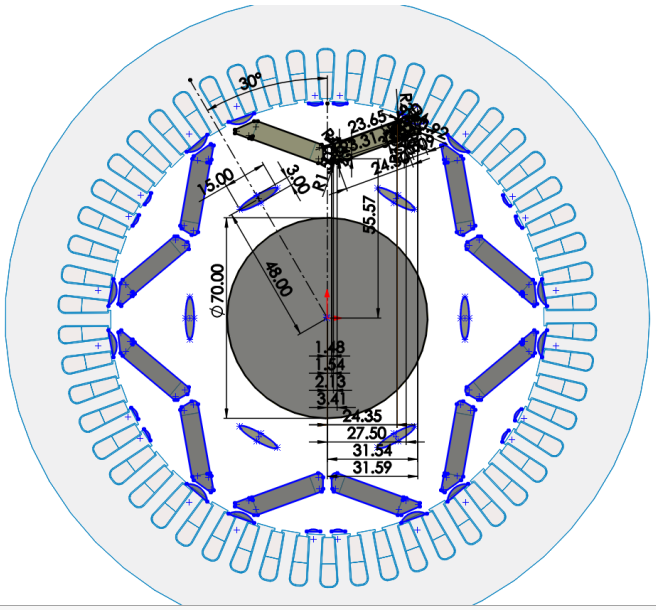


Figure 3.11: Tesla rotor dimensioning and design

3.3.4 Armature winding

Pole pitch

The pole pitch correspond to the pole arc, and it's expressed in meters by :

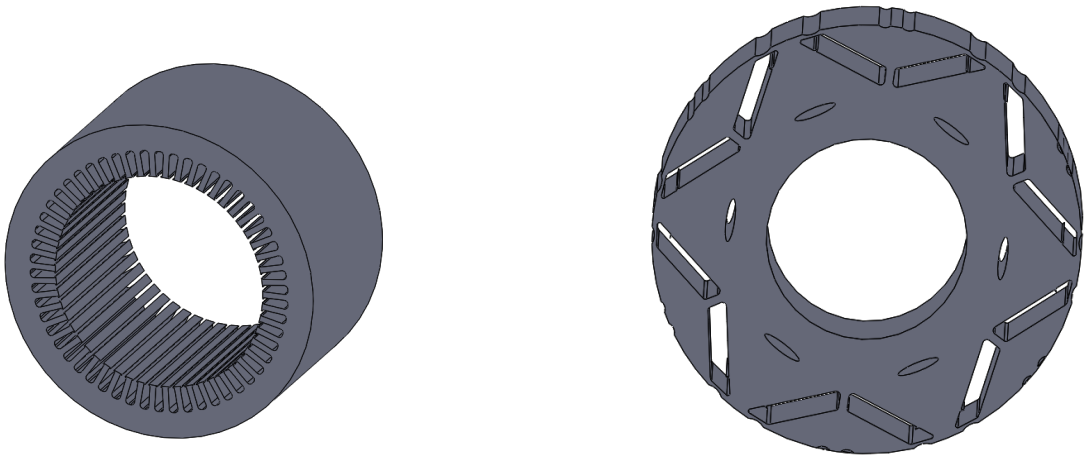
$$\tau_p = \frac{\pi D}{2p} \tag{3.1}$$

Where p is the number of pole pairs.

Phase zone

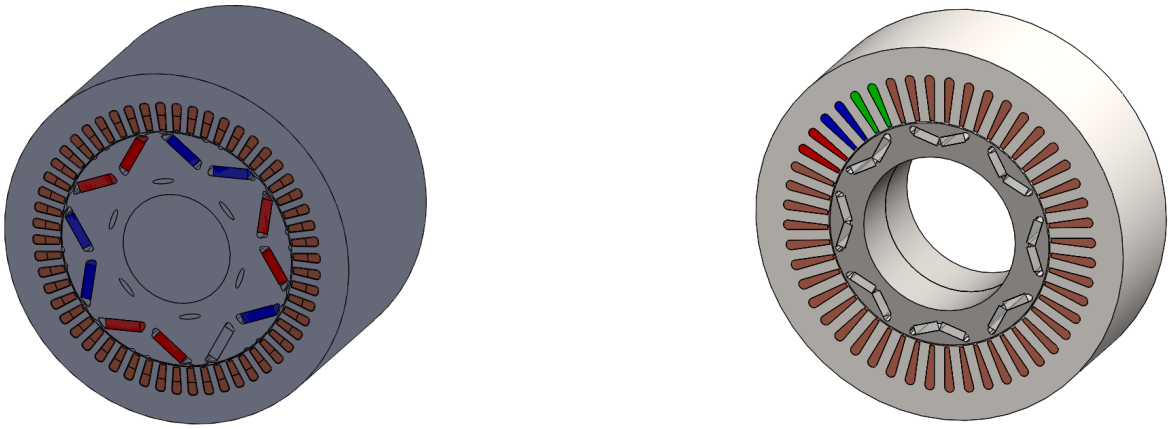
The phase zone distribution is written as:

$$\tau_v = \frac{\tau_p}{m} \tag{3.2}$$



(a) Tesla stator CAD model (b) Tesla rotor model

Figure 3.12: Tesla IPMSM Stator and rotor CAD models



(a) Tesla motor CAD model (b) Toyota Motor model

Figure 3.13: Tesla and Toyota IPMSM m CAD models

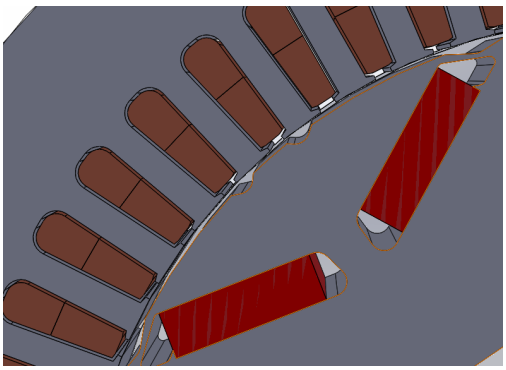
where m is a number of phases, this expresses the phase arc in meters per pole. the number of zone is $2pm$.

Number of slots per pole and phase

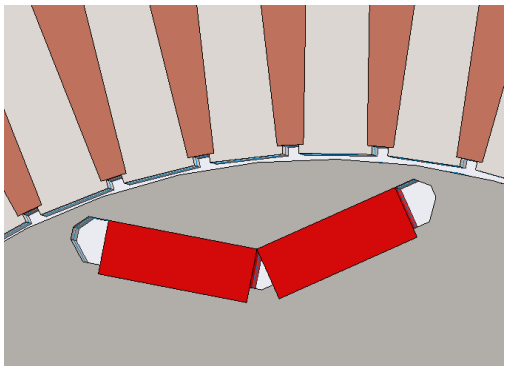
The number of slots for each zone is expressed by the term q, as a number of slots per pole and phase.

$$q = \frac{Q}{2pm} \tag{3.3}$$

where Q is the number of slots in the stator.



(a) Tesla pole pitch



(b) Toyota pole pitch

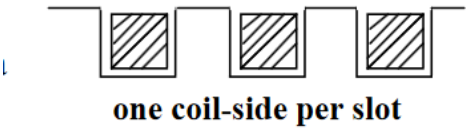
Figure 3.14: Pole pitch and slot zoom

Motor	Tesla Model 3	Toyota Prius
Pole pitch	0.078487m	0.063028
phase zone	0.02616m	0.02100
Number of slots per pole and phase	3	2

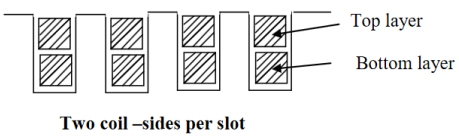
Table 3.8: Pole pitch and slots per pole phase

Winding

We are just interested in a full pitch single layer and double layer winding types. The Tesla motor winding is a double layer distributed winding and for the Toyota Prius the winding is a single layer distributed winding.

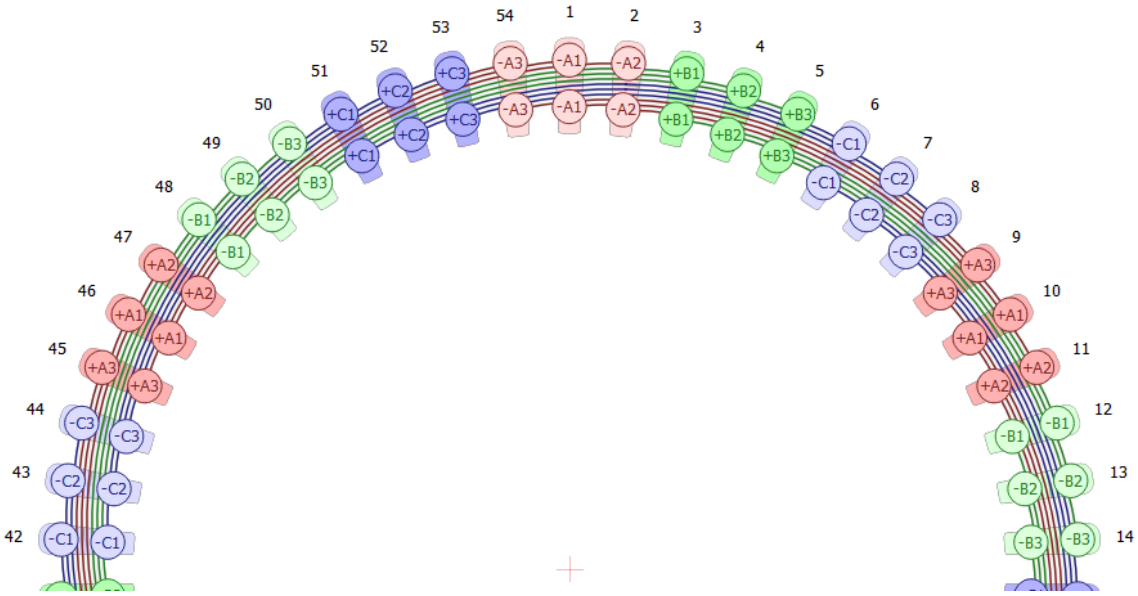


(a) Single layer winding

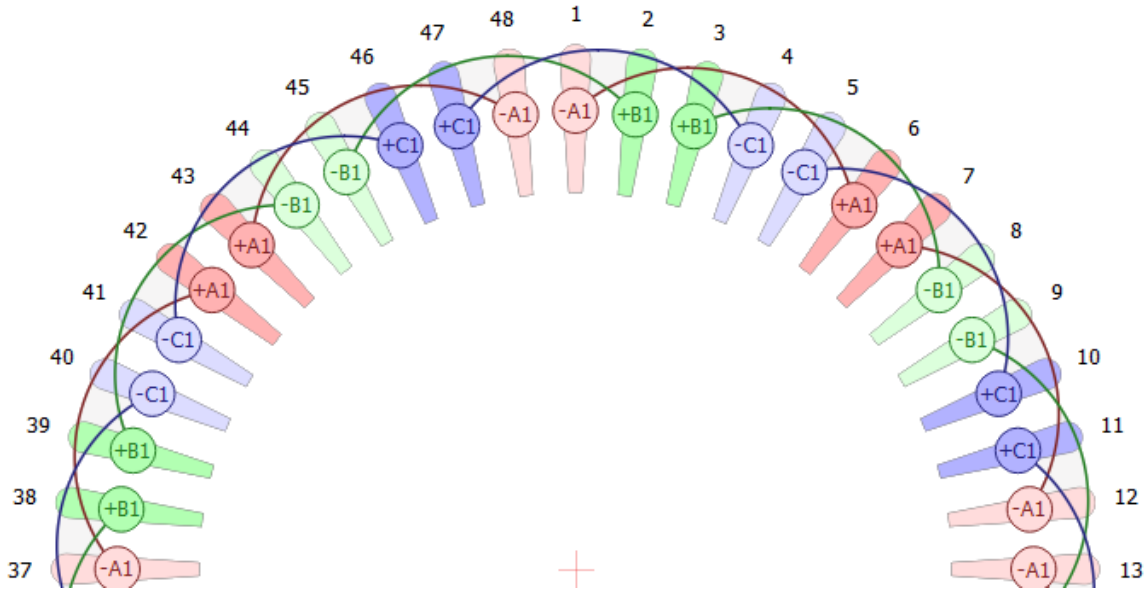


(b) Double layer winding

Figure 3.15: Single and double Layer winding



(a) Tesla winding /motorxp designstudio



(b) Toyota IPMSM winding /motorxp designstudio

Figure 3.16: Toyota and Tesla winding layout

Parallel paths and strands in hand

brushless permanent magnets are generally high speed motors, thus parallel paths are needed, for high current machines these parallel paths maybe within the coil itself and often referred to as "strands-in-hand."[18] Tesla uses 2 layers, 3 parallel paths, and 2 turns per coil,

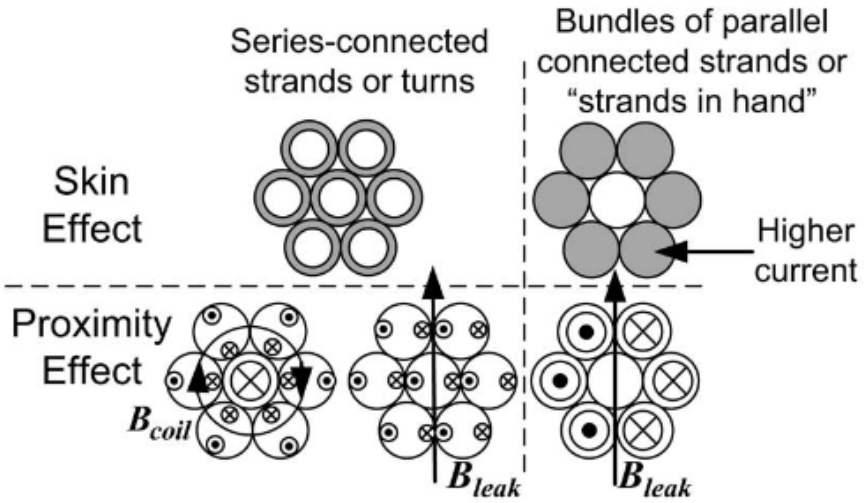


Figure 3.17: Skin and proximity effects in bundled coils with strands-in-hand

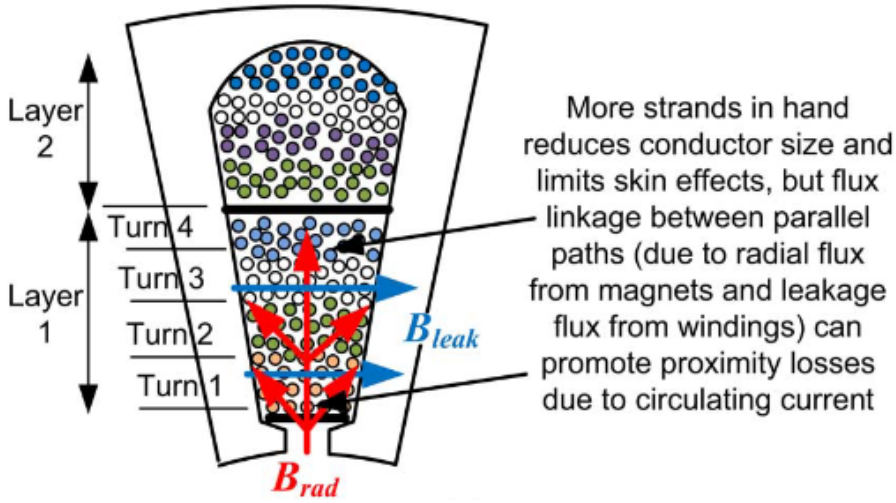


Figure 3.18: shows winding arrangement for a typical double layer winding with two coils sides per slot with four series turns per coil and 22 strands per turn (termed "strands-in-hand").

whereas the Toyota uses 13 strands in hands and 9 turns per coil.
The next figure shows the Tesla winding with 1 parallel paths,2 and 3 parallel paths.

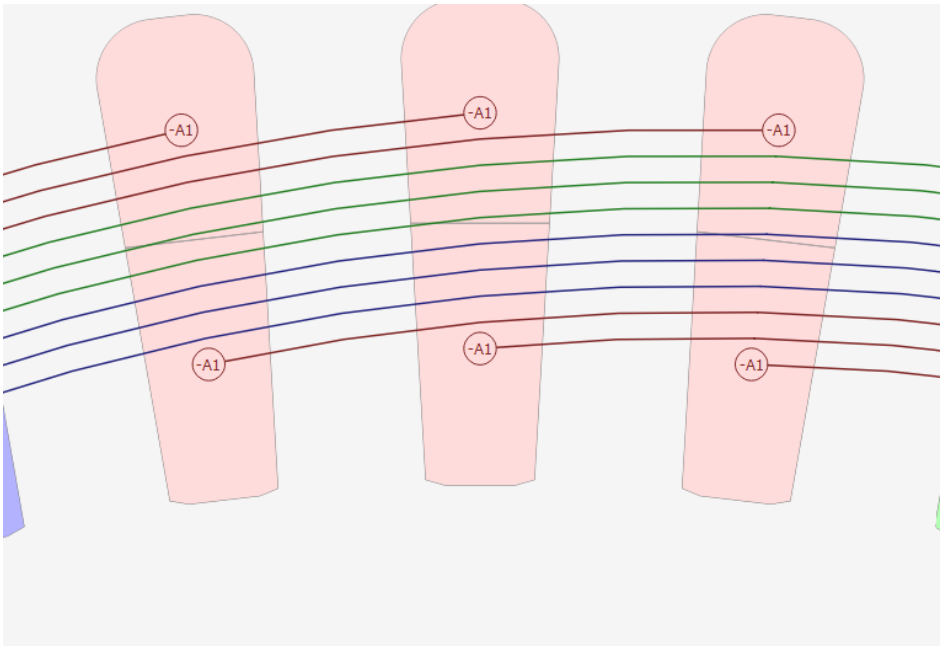


Figure 3.19: 1 parallel path winding

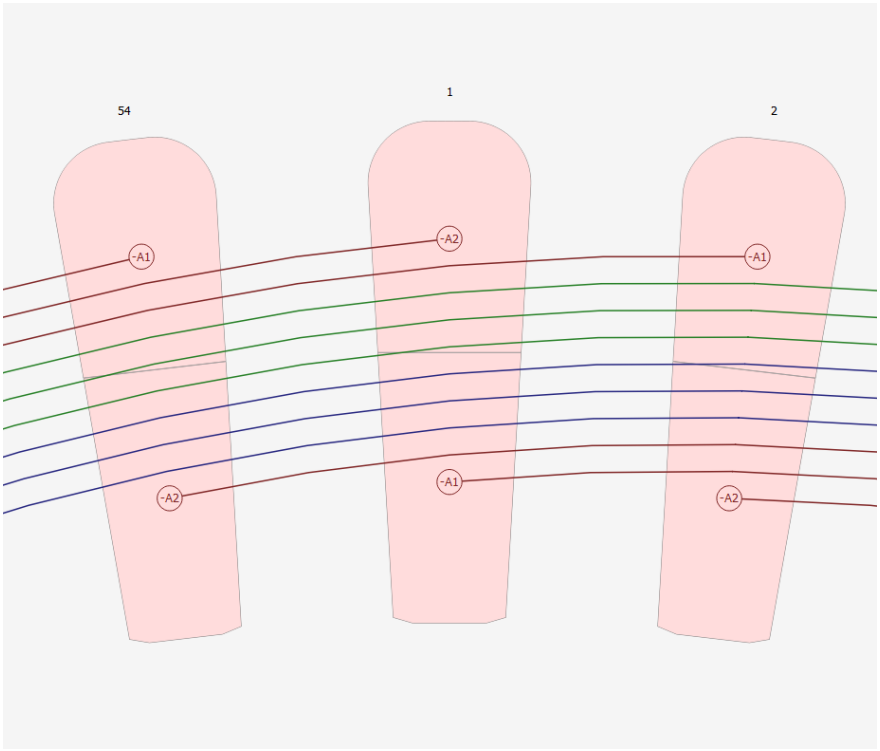


Figure 3.20: 2 parallel path winding

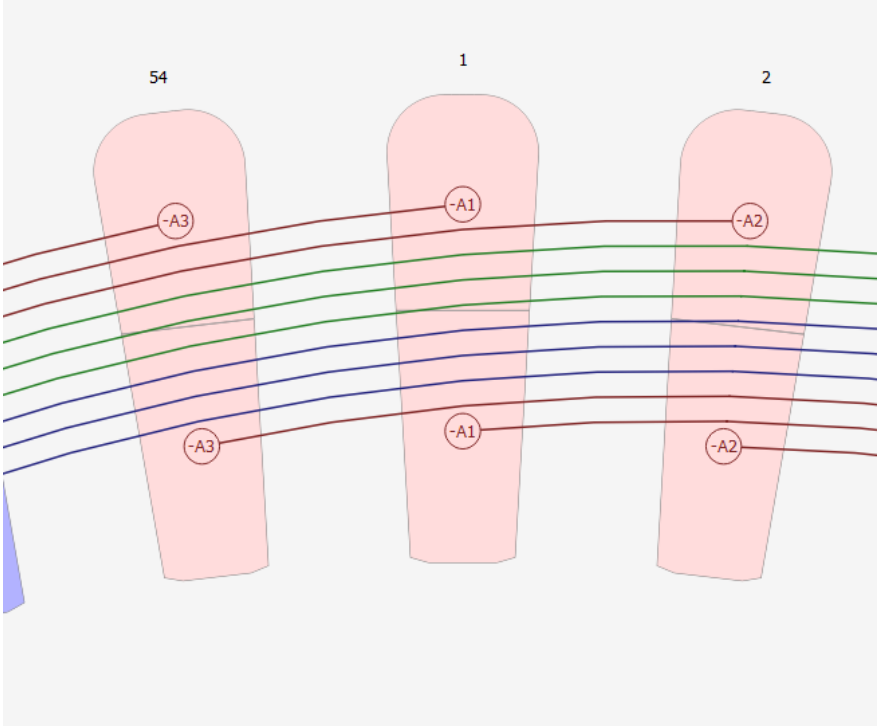


Figure 3.21: 3 parallel path winding

3.4 Simulation results comparison

Since we have demonstrated how powerful EMWorks 2D in simulating axial symmetrical models, we are going to use a 2D simulation and compare the results

3.4.1 No Load Analysis

The no load analysis results are obtained by setting the stator currents to 0, There are mainly the cogging torque which is the torque generated by the interaction between the ferromagnetic core and the permanent magnets. And the back EMF is obtained by running the motor in generator mode with an open circuit stator.

Cogging torque

We can see that the Tesla motor contains higher amplitude cogging Torque than the Toyota Prius, this is a result of how magnet flux interact with the stator teeth. In most applications, cogging torque is undesirable. The flux densities of both machines is presented below, we can see that the core material is not saturated by magnets, this mainly due to the optimized design.

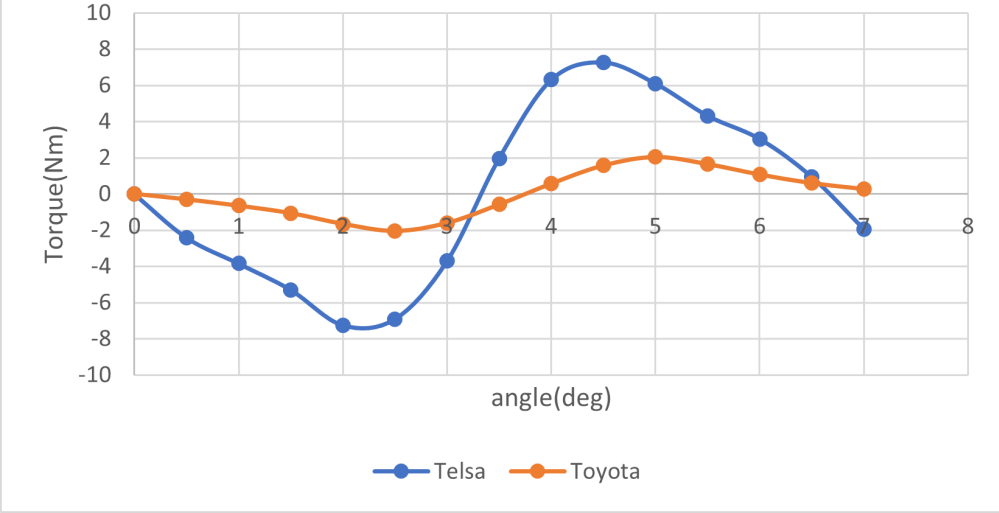
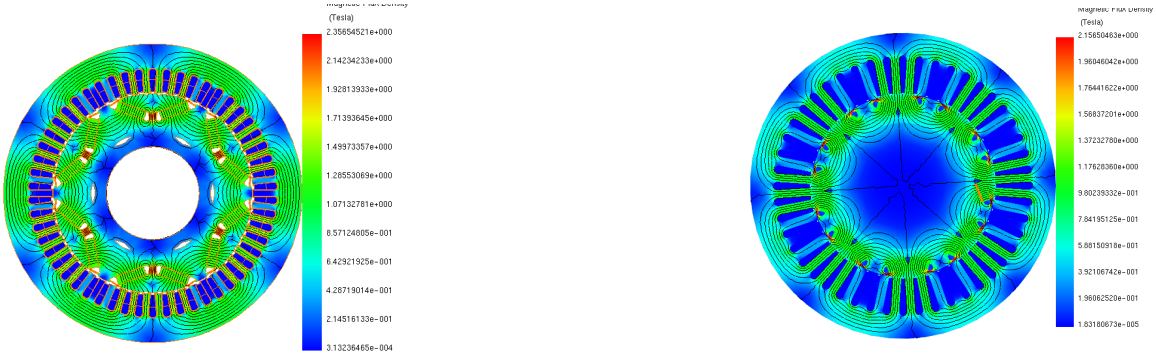


Figure 3.22: Tesla versus Toyota cogging Torque



(a) Tesla motor no load flux density

(b) Toyota motor no load flux density

Figure 3.23: No load flux density

Back emf

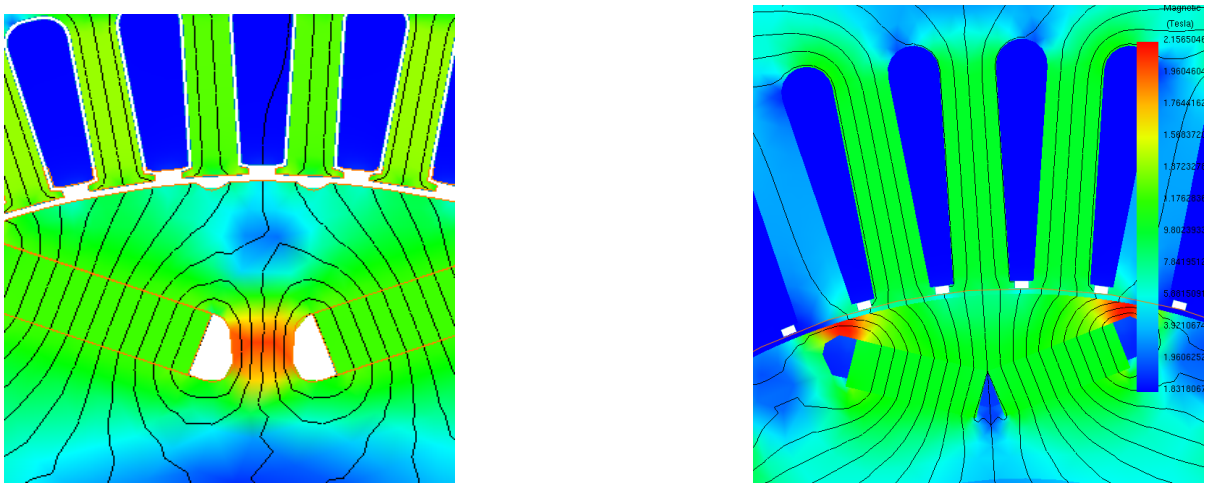
The back Electromotive Force is the induced voltage : The electrical and mechanical measures relation can be expressed by :

$$\theta_e = N_p \theta_m \tag{3.4}$$

Where p is the number of pair pole, θ_e is the electric angle and $\theta_m$is the mechanical angle. from this equation, we can derive the mechanical speed and electric frequency relation :

$$f_e = \frac{N_p S}{60} \tag{3.5}$$

Where S is the mechanical speed in RPM. by running the rotor at 1200RPM the frequency is 80Hz, the frequency is not needed on no load study.



(a) Tesla flux carrier zoom

(b) Toyota flux carrier zoom

Figure 3.24: Saturations on flux carriers

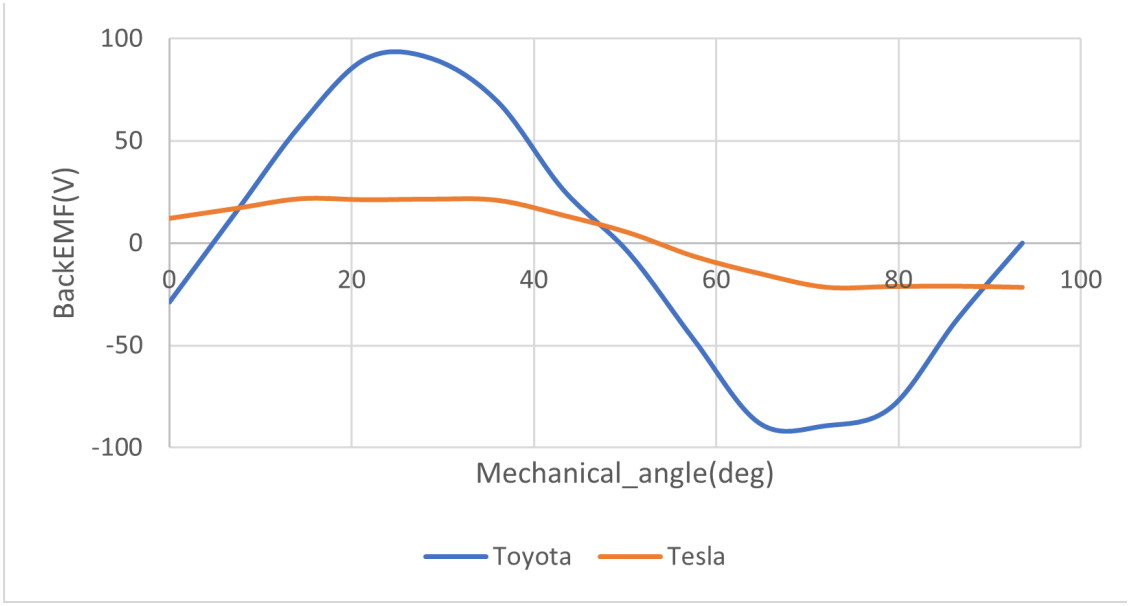


Figure 3.25: Tesla versus Toyota Back EMF at 1200RPM

From this figure we could see that the Back EMF of the Toyota is approximately sinusoidal, so it does not contain a lot of harmonics, The back EMF of the Tesla is approximately Trapezoidal means it has high order harmonic then the Toyota. if the back EMF peak value is high then it will cause high losses in the stator core, rotor core and permanent magnets at high speed.

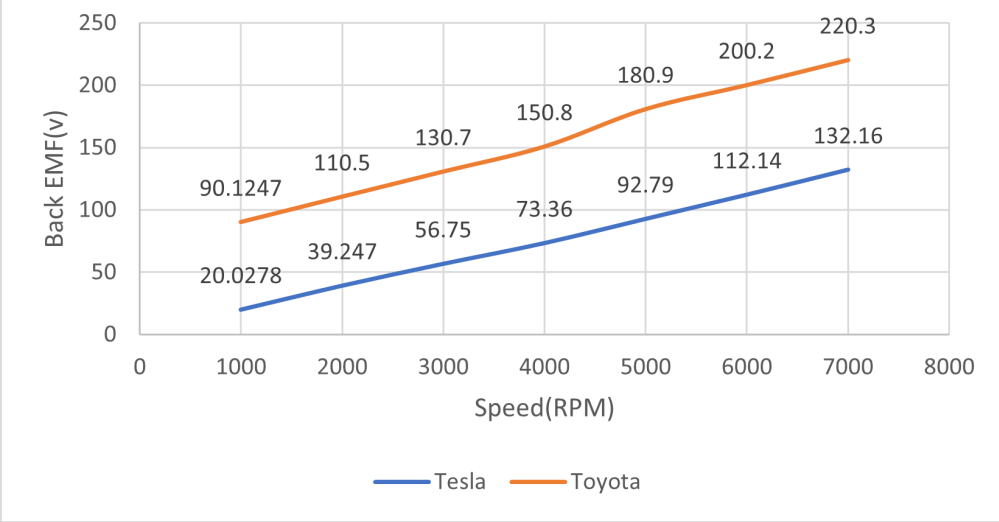


Figure 3.26: Tesla versus Toyota Back EMF with respect to speed

3.4.2 Load analysis

As we said in chapter 2 the stator self inductances are maximum when the rotor q-axis is aligned with the phase axis, while mutual inductances maximum when the rotor q-axis is in the midway between two phases. from the torque equation :

$$T_e = P * T_m = P * \frac{P_0}{\omega_s} = \frac{3P}{2}[\lambda_m I_q + (L_d - L_q) I_d I_q] \tag{3.6}$$

we can change the $(L_d - L_q) I_d I_q$ term by affecting the direct mutual inductance, one way to do that is to place the phase A with the D axis of the rotor and search for the electrical advance angle that gives the maximum starting Torque.

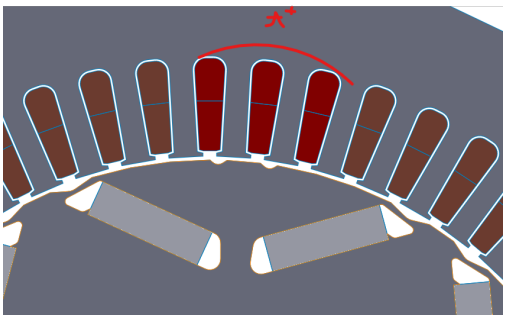


Figure 3.27: PhaseA+ centered on the rotor D-axis,

The load Torque of both motors is presented below:

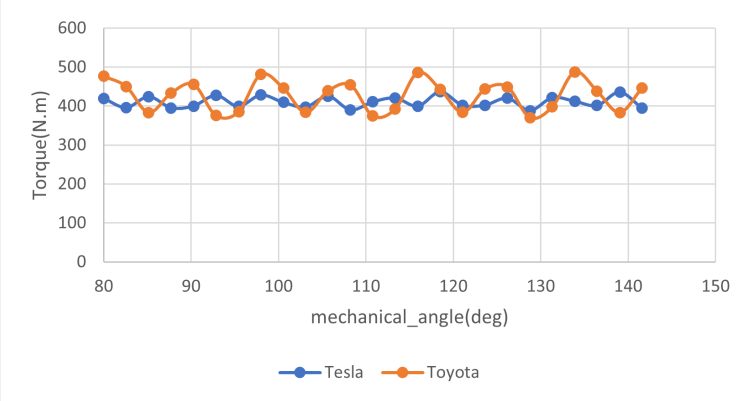


Figure 3.28: Load Torque

We could see that the torque ripple is higher in the Toyota Prius than the Tesla model 3 this was previewed by the back EMF results. To find The advance angle a bench of simulations is done using at each attempt a different advance angle.

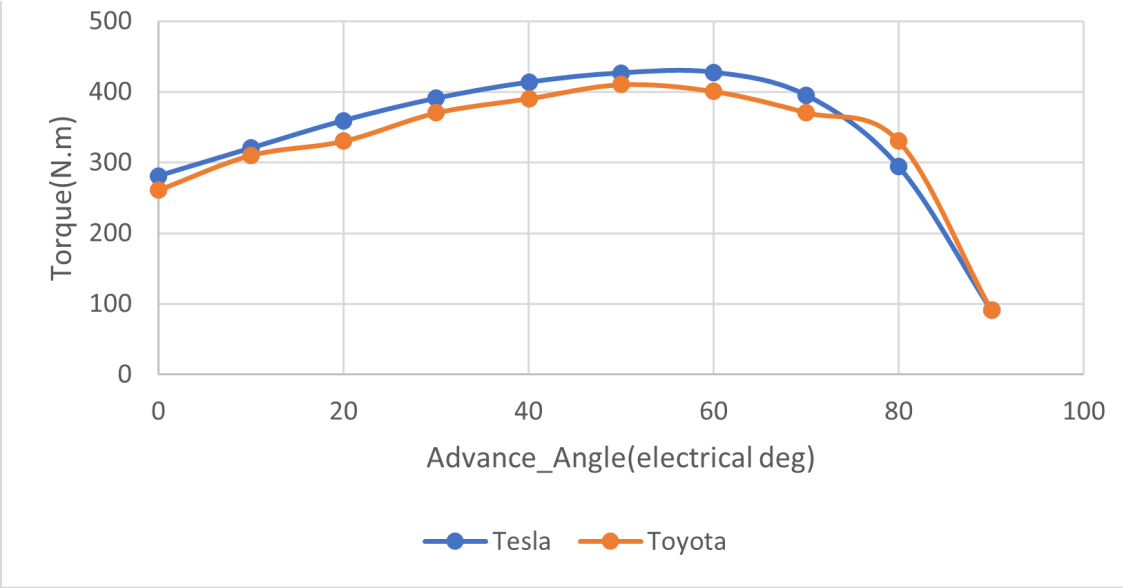


Figure 3.29: Torque with respect to advance angle

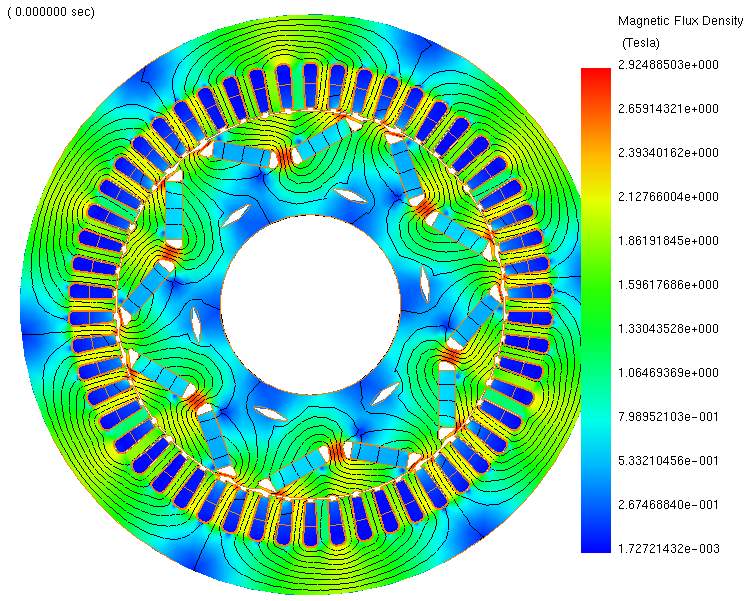


Figure 3.30: Tesla IPMSM flux density under load

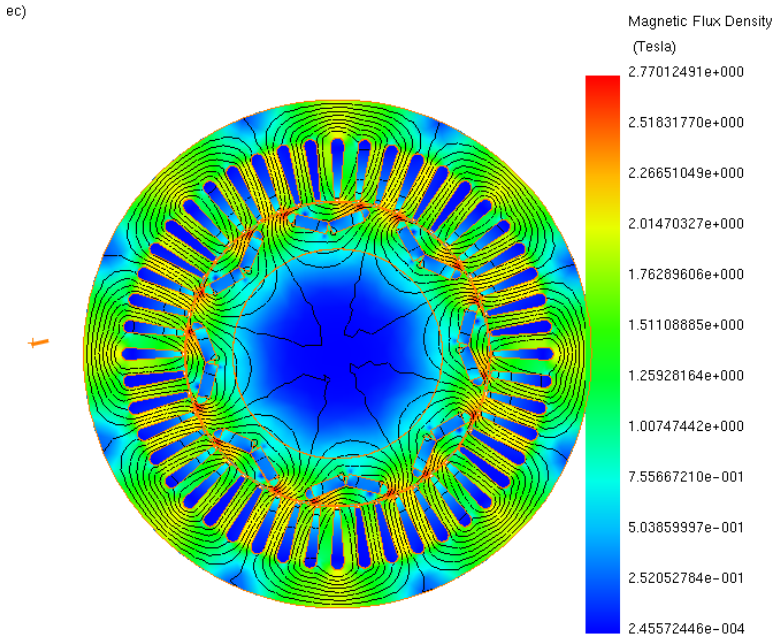
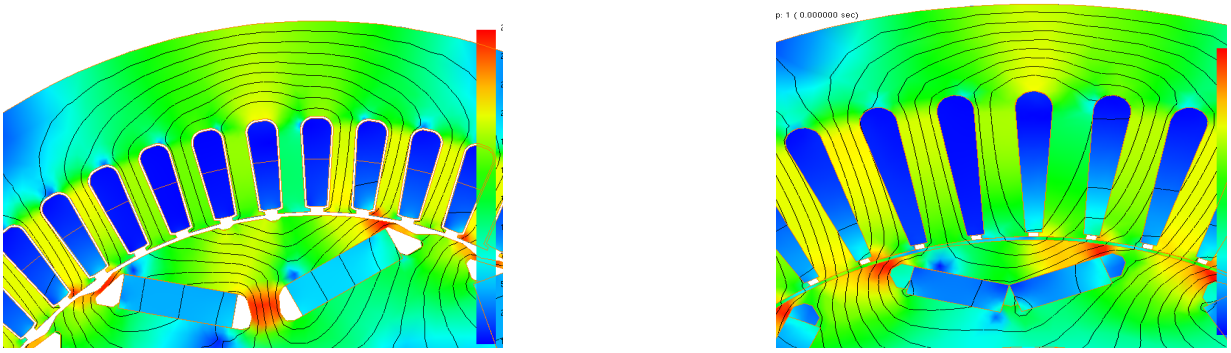


Figure 3.31: Toyota IPMSM flux density under load



(a) Tesla IPMSM flux zoom

(b) Toyota IPMSM flux zoom

Figure 3.32: Load Flux density on Toyota motor

From those figures, we could see that the phase axis is approximately aligned with rotor pole d-axis, a bit of material saturation could be seen on the edge of the poles and on the flux carriers. The Torque value given by this model is 412Nm for the Tesla approximately 420Nm as the experimental value and 390Nm for the Toyota motor, which means that the simulation yields good results.

3.5 Changing rotor geometry and materials

3.5.1 Changing rotor geometry

Rotor air cavity removal effects

In this section, we will be studying how changing rotor geometry will affect motor torque and material saturation.



(a) Rotor before removing cavities

(b) Rotor after removing cavities

Figure 3.33: Load flux density on Toyota IPMSM

No load study

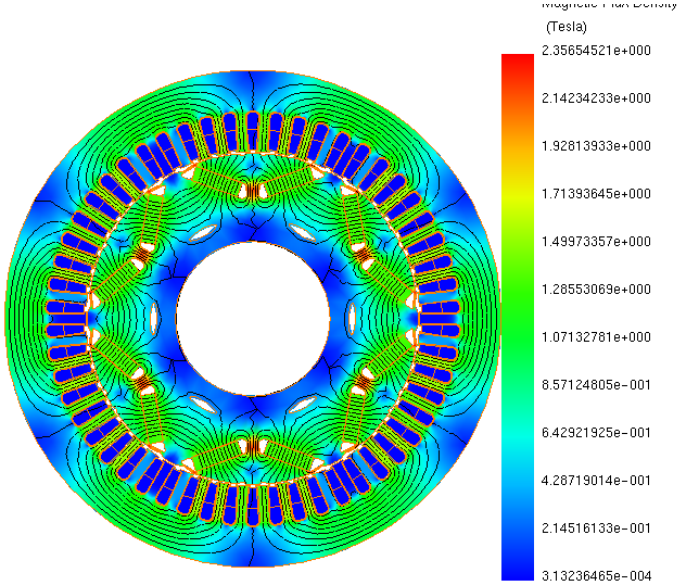


Figure 3.34: Magnet flux before cavity removal

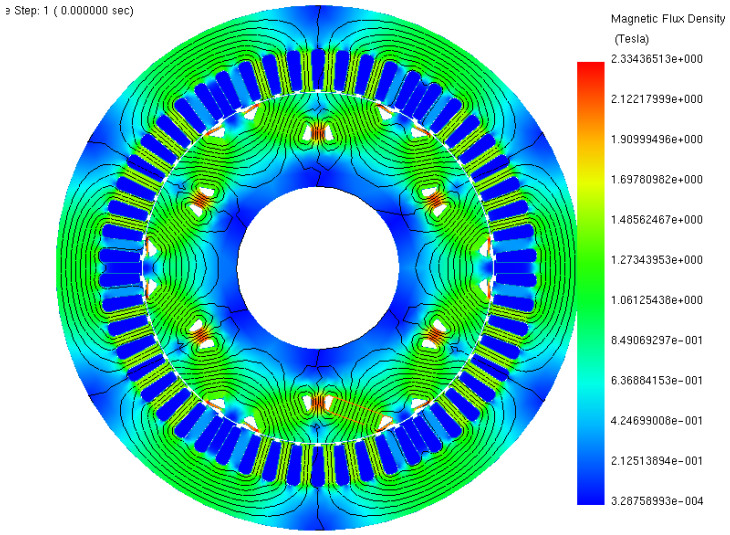


Figure 3.35: Magnet flux after cavity removal

From this figure, we could see that the flux density is approximately the same, however the removal can affect the cogging torque, the next figure shows the cogging torque before and after cavity removal.

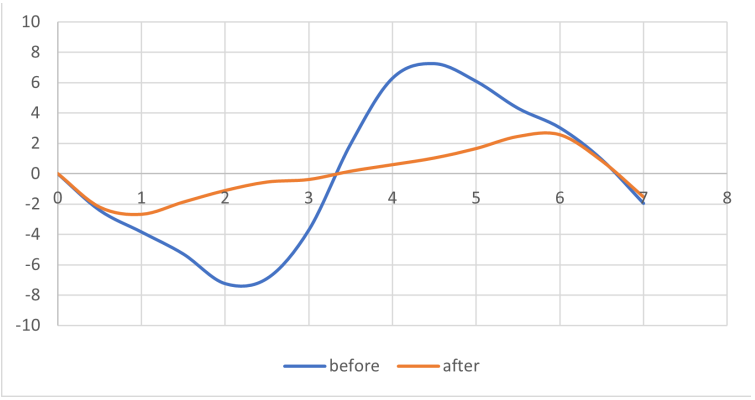


Figure 3.36: Cogging torque after and before geometry modification

we could see that the cogging toque is lower, since cogging torque is generally undesirable this would be more effective, but before judging that we have to compare the load torque results. Some interesting notice is that the interior region of the rotor contains a very low flux density this could prove why Tesla engineers used those cavities, for the purpose of mass minimization since it does not affect the overall torque, the next study proves this fact.

Load analysis

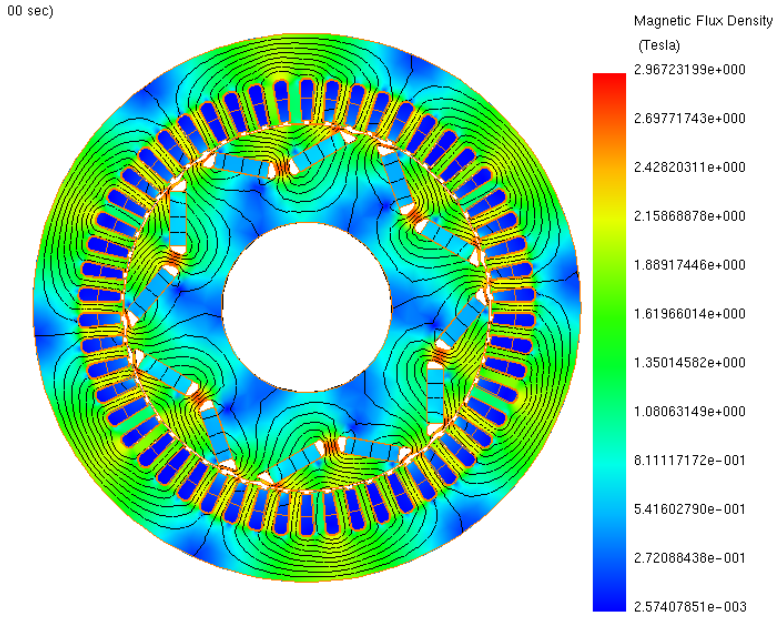


Figure 3.37: Flux density of Tesla IPMSM after cavity removal

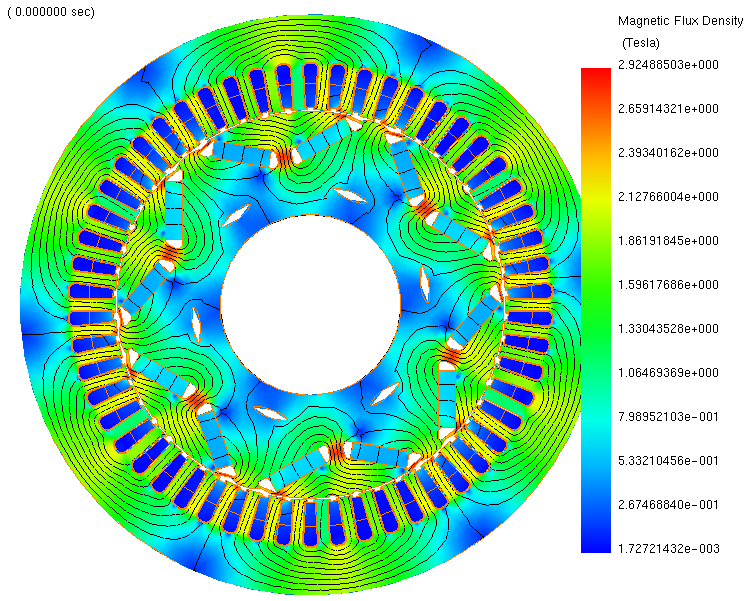


Figure 3.38: Flux density of Tesla IPMSM before cavity removal

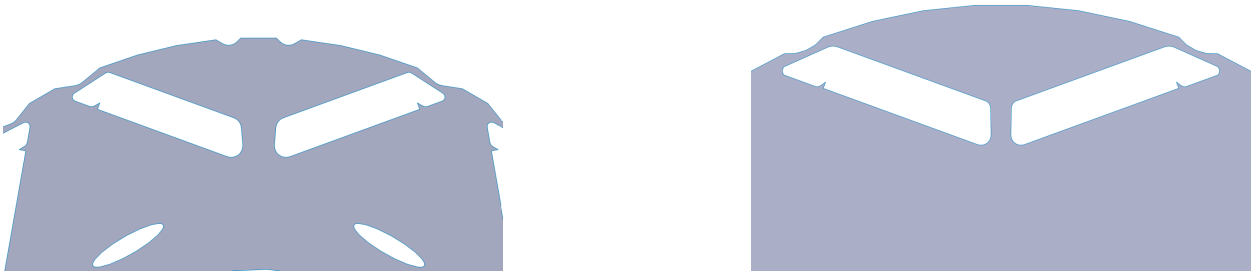


Figure 3.39: Load torque after and before cavity removal

It's clear that the results are approximately the same, in addition, the blue areas that contain low flux density are the regions where we can insert cavities, thus reducing rotor weight, which in turns will optimize the performance. Next we will be changing the V-shaped magnet cavity dimensions with keeping the real magnet dimensions and the angle

will hold the same number of turns per coil, and see it's effects.

3.5.2 Changing the Vshaped rotor cavity dimensions



(a) Rotor before modifying the V-shaped cavity

(b) Rotor after modifying the V-shaped cavity

Figure 3.40: Rotor before and after modification

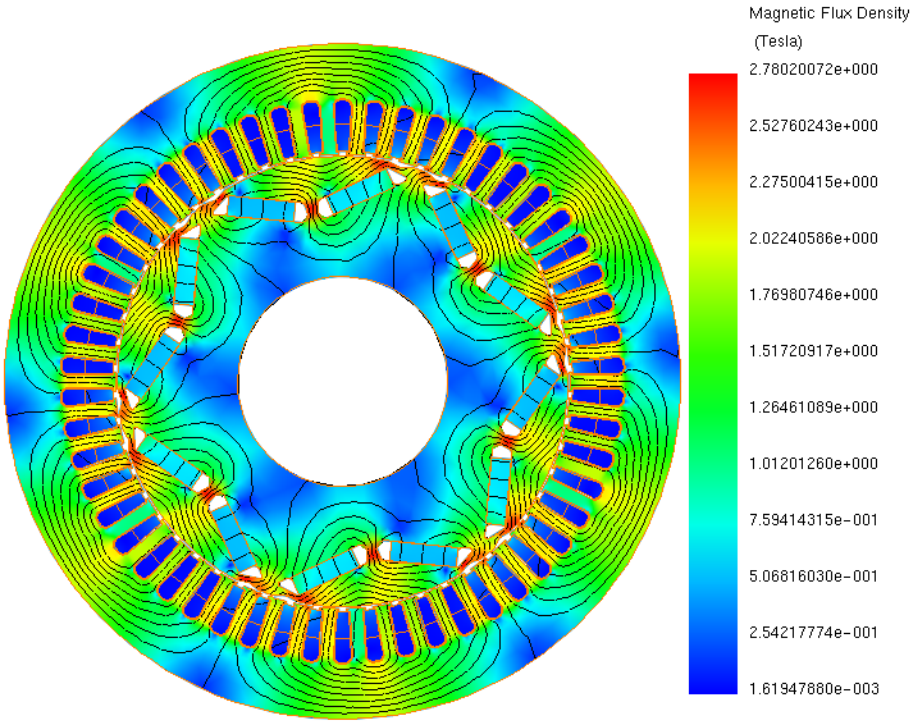


Figure 3.41: Load flux density with changed V-shaped cavity

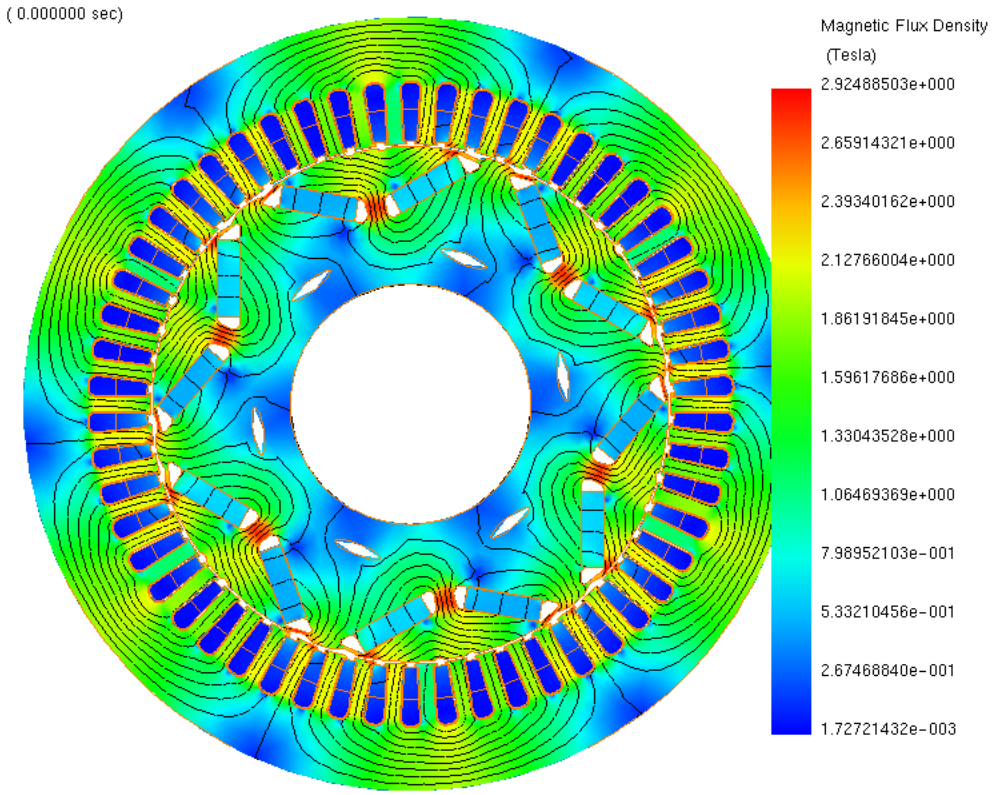


Figure 3.42: Flux density before change

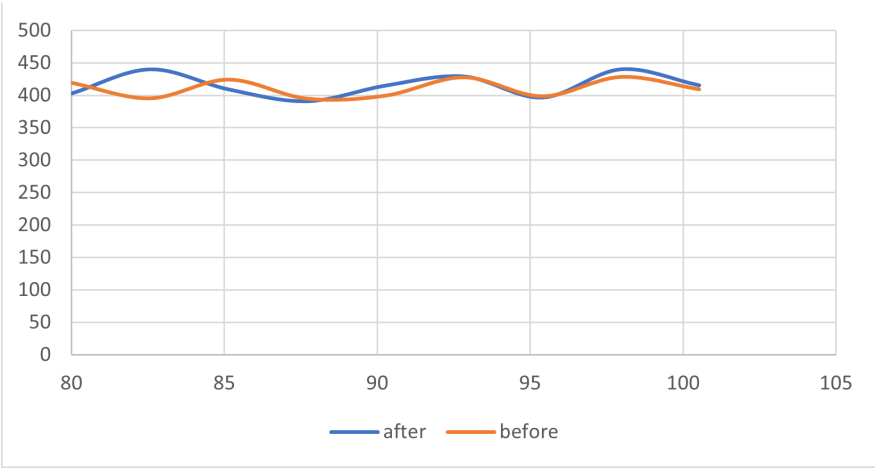


Figure 3.43: Load torque after and before changing V-Shaped cavity

From this figure we can see that the results are better with the new dimensions, the

magnets dimension have been kept as the previous model, and we can conclude that the new dimensions are more optimized. The new average value of Tesla torque is 417Nm means higher by 5Nm from the previous model

3.5.3 N52 vs N30H load torque results

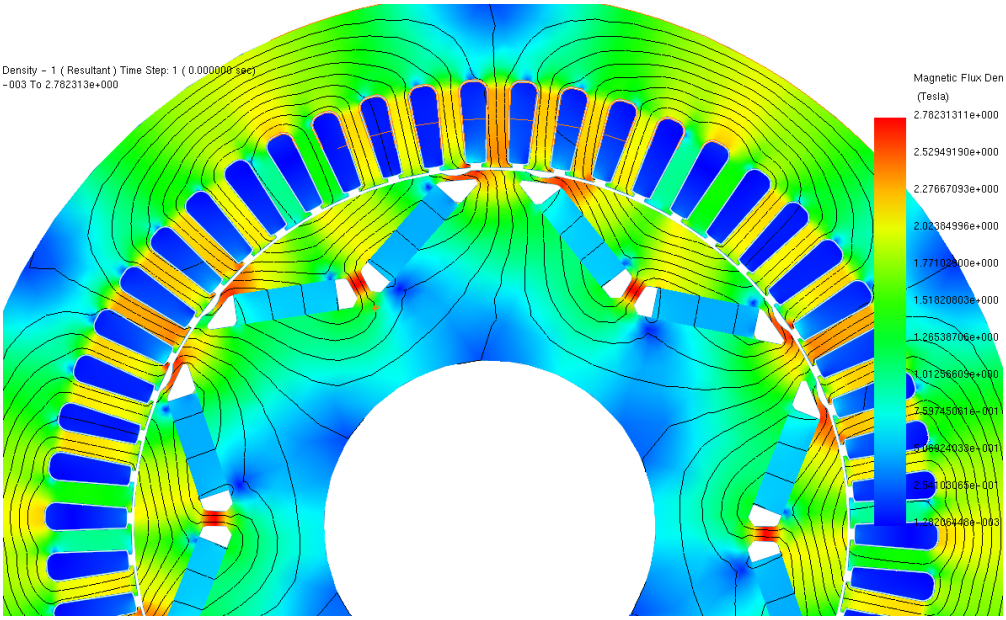


Figure 3.44: Load flux density using N30

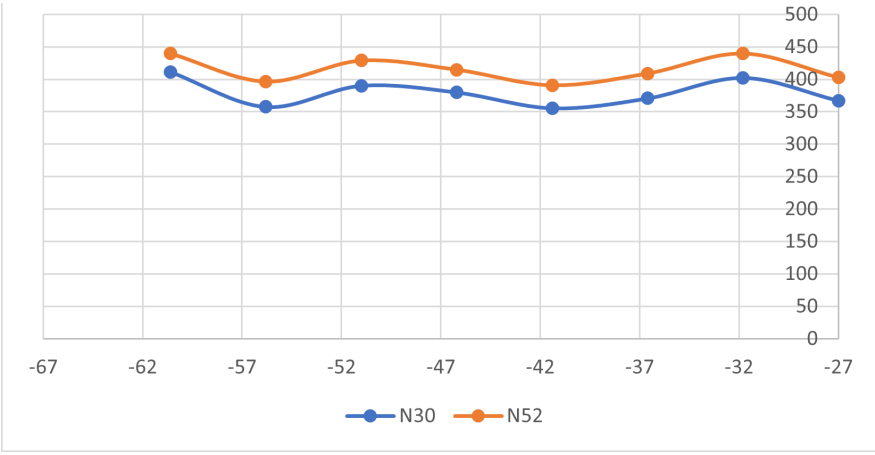


Figure 3.45: Load Torque using N30H and N52 magnets

We conclude that changing magnetic material to a lower grade affects the total torque, which means choosing the type of magnet material is a preselection step with respect to the

desired torque, and an experimental or a simulation study must be carried out to prove the material selection.

3.6 Conclusion

In this chapter, we have compared 3D versus 2D simulation results of an axial symmetric model, we have also designed simplified models using SolidWorks of the Tesla Model 3 and the Toyota Prius motors ,and we have simulated and evaluated their performance, we have also checked the effect of geometry modification on performance and demonstrated how we could use FEA results to optimize our models.

Chapter 4

Conclusion and prospects

Along this report, we presented the different steps of modeling and simulation of IPMSM . We started the report by an overview about the company by stating the company clients and products.
In chapter 2 we have invoked the types of electric motors that have been and are used in electric cars, then we presented the d-q model and phasor diagram of the IPMSM which demonstrated that those motors are a hybrid combination between two motors, that are the reluctance synchronous motor and the conventional permanent magnet synchronous motor, we determined the torque equation using the direct and quadrature axis currents and inductances since it's important to understand how we can get the maximum torque for those motors .we have also briefly presented the FEA equations and methods for resolving electromagnetic field equations.
After that we regrouped the results of the numerical study in chapter 3,by starting with the 2D versus 3D FEA analysis of an PMSM motor under no load, and we have demonstrated how useful using 2D model instead of a 3D model when the system is symmetrical.Then we have presented the Toyota Prius 2004 and the Tesla Model 3 2018 IPMSM motors characteristics and dimensions ,and we have modeled those motors using SolidWorks as a CAD environment after that we have simulated those models using EMWorks2D and compared the results. finely, We have also studied the effects of geometry and magnet materials on performance, and we found a better model that gave us better result. With this study and discussion, we have demonstrated the importance of CAD and simulation tools that can be used to understand and optimize complex systems such as electric motors by using FEA method. Last but not least, some numerical studies that have not been conducted throughout this work are losses computations, since losses needs a 3D model to be determined and due to the insufficient computational resource that we have on our computers we could not calculate losses, by calculating losses we could then calculate the efficiency of the motor.This could be accomplished in future work.

Appendix A

Finite element applied to a 2D problem

We will be taking the time harmonic form of the diffusion equation with A, the z component of the magnetic vector potential A as the unknown, For a linear 2D Cartesian problem :

$$\frac{1}{\mu}\frac{\partial^2 A}{\partial^2 x} + \frac{1}{\mu}\frac{\partial^2 A}{\partial^2 y} = -J_0 + j\omega\sigma A \tag{A.1}$$

which is a general form of the Poisson equation presented in chapter 2 equation 2.30
ω is the angular velocity
σ is the electrical conductivity
μ is the magnetic permeability

Using an approximation of A,$\hat{A}$, gives as a residual as described in chapter 2:

$$R = \frac{1}{\mu}\frac{\partial^2 \hat{A}}{\partial^2 x} + \frac{1}{\mu}\frac{\partial^2 \hat{A}}{\partial^2 y} - j\omega\sigma\hat{A} + J_0 \tag{A.2}$$

Multiplying by a weighting function and setting the integral to zero

$$\int_\Omega RW dxdy = 0 \tag{A.3}$$

$\Omega$is the surface on which we are calculating.

Substituting for R:

$$-\iint_\Omega W(\frac{1}{\mu}\frac{\partial^2 \hat{A}}{\partial^2 x} + \frac{1}{\mu}\frac{\partial^2 \hat{A}}{\partial^2 y})dxdy + j\omega\sigma\iint_\Omega W\hat{A}dxdy = \iint_\Omega W J_0 dxdy \tag{A.4}$$

Now we can integrate the first term by parts using Green's second identity.

This identity is derived from the divergence theorem applied to a vector field $F = \psi\nabla\phi$

$$\int_U (\psi\nabla^2\phi - \nabla\phi\nabla\psi)dV = \oint_{\partial U}(\psi\frac{\partial\phi}{\partial n} - \phi\frac{\partial\psi}{\partial n})dS \tag{A.5}$$

Where :
$\frac{\partial \phi}{\partial n}$ is the directional derivative of ϕin the direction of the outward pointing surface normal n.[19].

using this on first term of equation A.4 we obtain :

$$-\iint_{\Omega} W(\frac{1}{\mu}\frac{\partial^2 \hat{A}}{\partial^2 x}+\frac{1}{\mu}\frac{\partial^2 \hat{A}}{\partial^2 y})dxdy=\iint_{\Omega}\frac{1}{\mu}(\frac{\partial W}{\partial x}\frac{\partial \hat{A}}{\partial x}+\frac{\partial W}{\partial y}\frac{\partial \hat{A}}{\partial y})dxdy-\oint_{C}\frac{1}{\mu}W\frac{\partial \hat{A}}{\partial n}dc \quad (A.6)$$

Where C is the boundary of Ω and n is the normal vector to Ω.

we now break the surface integrals into summations over small areas. In the case of EMS and EMWorks 2D, the surface will be meshed with triangles (the finite elements) and we will replace the integral over the entire domain with the summation of the integral over the individual triangles.
substituting in A.4 we get :

$$\sum_{M}\frac{1}{\mu^e}\iint_{\Omega^e}(\frac{\partial W^e}{\partial x}\frac{\partial A^e}{\partial x}+\frac{\partial W}{\partial y}\frac{\partial A^e}{\partial y})dxdy-\frac{1}{\mu^e}\frac{\partial A^e}{\partial n}\oint_{C}W^e dc+j\omega^e\sigma\iint_{\Omega^e}W^e A^e dxdy=J_0\iint_{\Omega^e}W^e dxdy \quad (A.7)$$

Where M is the number of triangle elements
The line integral in equation 1.8 only needs to be evaluated over elements which have a side in common with the boundary of the problem. Normally this integral is simply set to zero implying that $\frac{\partial A}{\partial n}=0$ as a boundary condition, stated in chapter 2.

Consider a triangular element (ijk) depicted in Figure A.1 We will adopt the convention of counter clockwise numbering of the vertices. The vertices are nodes at which the unknown vector potentials will eventually be calculated.

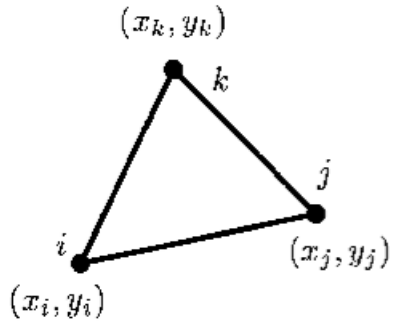


Figure A.1: mesh element

If we now assume that the potential varies linearly in the element, we obtain what is known as a linear or first order element. With this approximation, we may express the vector potential at any point in the triangle as:

$$\hat{A}=C_1+C_2x+C_3y \quad (A.8)$$

where $C_1, C_2, C_3$are constant to be determined. Note that since the vector potential varies linearly, the flux density, which is the derivative of the potential, is constant in the triangle. At vertex i, we have $x = x_i$,and $y = y_i$at this point $\hat{A}$ must be equal to$\hat{A}_i$ and we get :

$$C_1 + C_2 x_i + C_3 y_i = \hat{A}_i \tag{A.9}$$

$$C_1 + C_2 x_j + C_3 y_j = \hat{A}_j \tag{A.10}$$

$$C_1 + C_2 x_k + C_3 y_k = \hat{A}_k \tag{A.11}$$

his gives three equations and three unknowns. We now solve for C_1, C_2 and C_3 in terms of the potential and geometry. Using Kramer's rule:

$$C_1 = \frac{\begin{vmatrix} \hat{A}_i & x_i & y_i \\ \hat{A}_j & x_j & y_j \\ \hat{A}_k & x_k & y_k \end{vmatrix}}{\begin{vmatrix} 1 & x_i & y_i \\ 1 & x_j & y_j \\ 1 & x_k & y_k \end{vmatrix}} \tag{A.12}$$

$$C_1 = \frac{\begin{vmatrix} \hat{A}_i & x_i & y_i \\ \hat{A}_j & x_j & y_j \\ \hat{A}_k & x_k & y_k \end{vmatrix}}{2\Delta} \tag{A.13}$$

Where Δ is the area of the triangle
same calculations for C_2 and C_3

using those results, we may now express $\hat{A}$as :

$$\hat{A} = \frac{(a_i + b_i x + c_i y)\hat{A}_i + (a_j + b_j x + c_j y)\hat{A}_j + (a_k + b_k x + c_k y)\hat{A}_k}{2\Delta} \tag{A.14}$$

Where : $a_i = x_j y_k - x_k y_j$
$bi = y_j - y_k$
$c_i = x_k - x_j$
The coefficients of the nodal potentials in equation A.14 are called shape functions. The potential can be expressed as the sum of the shape functions times the nodal potential.

$$\hat{A} = \sum_{i=1}^{m} N_i(x, y)\hat{A}_i \tag{A.15}$$

Where there are m nodes in the element and N_i are the shape functions. These functions have the following important properties:

1.$N_i = 1$ at node i and 0at all other nodes
2.The sum of all the shape function at any point in the triangle is equal to 1.
Writing the nodal potential in the element in matrix form, since equation A.15 it can be seen as scalar product

$$\hat{A}^e = \begin{pmatrix} N_i^e & N_j^e & N_k^e \end{pmatrix} \begin{pmatrix} \hat{A_i}e \\ \hat{A}_j^e \\ \hat{A}_k^e \end{pmatrix} \tag{A.16}$$

Where :

$$N_i^e = \frac{(a_i^e + b_i^e x + c_i^e y)}{2\Delta} \tag{A.17}$$

$$N_j^e = \frac{(a_j^e + b_j^e x + c_j^e y)}{2\Delta} \tag{A.18}$$

$$N_k^e = \frac{(a_k^e + b_k^e x + c_k^e y)}{2\Delta} \tag{A.19}$$

In the Galerkin Method, we choose the weighting function to be the same as the shape function, as we said in chapter 2.

$$W^e = \begin{pmatrix} N_i^e \\ N_j^e \\ N_k^e \end{pmatrix} \tag{A.20}$$

Taking derivatives with respect to x and y:

$$\frac{\partial \hat{A}}{\partial x} = \frac{1}{2\Delta} \begin{pmatrix} b_i^e & b_j^e & b_k^e \end{pmatrix} \begin{pmatrix} \hat{A}_i \\ \hat{A}_j \\ \hat{A}_k \end{pmatrix} \tag{A.21}$$

$$\frac{\partial \hat{A}}{\partial y} = \frac{1}{2\Delta} \begin{pmatrix} c_i^e & c_j^e & c_k^e \end{pmatrix} \begin{pmatrix} \hat{A}_i \\ \hat{A}_j \\ \hat{A}_k \end{pmatrix} \tag{A.22}$$

and

$$\frac{\partial W^e}{\partial x} = \frac{1}{2\Delta} \begin{pmatrix} b_i^e \\ b_j^e \\ b_k^e \end{pmatrix} \tag{A.23}$$

$$\frac{\partial W^e}{\partial y} = \frac{1}{2\Delta} \begin{pmatrix} c_i^e \\ c_j^e \\ c_k^e \end{pmatrix} \tag{A.24}$$

Using these, the first term on the left side of Equation A.7 becomes:

$$\frac{1}{\mu^e}\left(\frac{\partial W^e}{\partial x}\frac{\partial \hat{A}}{\partial x} + \frac{\partial W^e}{\partial y}\frac{\partial \hat{A}}{\partial y}\right) \iint_{\Omega^e} dxdy \tag{A.25}$$

We recognize that :

$$\iint dxdy = \Delta \tag{A.26}$$

Substituting :

$$\frac{1}{\mu^e}\iint_{\Omega^e}(\frac{\partial W^e}{\partial x}\frac{\partial \hat{A}^e}{\partial x}+\frac{\partial W^e}{\partial y}\frac{\partial \hat{A}^e}{\partial y})dxdy = \frac{1}{4\mu^e\Delta}\begin{pmatrix} b_i^2+c_i^2 & b_ib_j+c_ic_j & b_ib_k+c_ic_k \\ b_ib_j+c_ic_j & b_j^2+c_j^2 & b_jb_k+c_jc_k \\ b_ib_k+c_ic_k & b_jb_k+c_jc_k & b_k^2+c_k^2 \end{pmatrix}\begin{pmatrix}\hat{A}_i\\ \hat{A}_j\\ \hat{A}_k\end{pmatrix} \tag{A.27}$$

The second integral in equation A.7:

$$j\omega\sigma^e\iint_{\Omega^e}W^eA^edxdy = j\omega\sigma^e\iint_{\Omega^e}\begin{pmatrix}N_i^e\\ N_j^e\\ N_k^e\end{pmatrix}\begin{pmatrix}N_i^e & N_j^e & N_k^e\end{pmatrix}\begin{pmatrix}\hat{A}_i\\ \hat{A}_j\\ \hat{A}_k\end{pmatrix}dxdy = \frac{j\omega\sigma^e\Delta}{12}\begin{pmatrix}2&1&1\\1&2&1\\1&1&2\end{pmatrix}\begin{pmatrix}\hat{A}_i\\ \hat{A}_j\\ \hat{A}_k\end{pmatrix} \tag{A.28}$$

The forcing function (right hand side of equation A.7) becomes:

$$J_0\iint_{\Omega^e}\frac{(a_i+b_ix+c_iy)}{2\Delta}dxdy = J_0\frac{(a_i+b_i\bar{x}+c_i\bar{y})}{2} \tag{A.29}$$

Where $\bar{x}$and $\bar{y}$are the coordinates of the triangle center.

$$\bar{x} = \frac{1}{3}(x_i+x_j+x_k) \tag{A.30}$$

$$\bar{y} = \frac{1}{3}(y_i+y_j+y_k) \tag{A.31}$$

Assuming that J_0 is constant over the triangle, we get:

$$J_0^e\iint_{\Omega^e}W^edxdy = \frac{J_0^e\Delta}{3}\begin{pmatrix}1\\1\\1\end{pmatrix} \tag{A.32}$$

Assembly

Once the element matrices are found for each element, they are used to form the global or system matrix in a process known as assembly [20].

Appendix B

Demagnetization Characteristics

ARNOLD®
MAGNETIC TECHNOLOGIES

Neodymium-Iron-Boron Magnet Grades

Summary Product List & Reference Guide

Basic Grades

	Properties	B_r		H_{cB}		H_{cJ}		$(BH)_{max}$		Temp. Coef.		T_w
	Grade**	Typical mT	Typical gauss	min kA/m	min oersteds	min kA/m	min oersteds	Typical kJ/m^3	Typical MGOe	$\alpha(B_r)$ %/°C	$\alpha(H_{cJ})$ %/°C	max °C
4	**N35**	1210	12100	860	10800	955	12000	282.5	35	-0.12	-0.618	80
5	**N38**	1260	12600	860	10800	955	12000	306	38	-0.12	-0.618	80
6	**N40**	1270	12700	923	11600	955	12000	318	40	-0.12	-0.618	80
7	**N42**	1315	13150	860	10800	955	12000	334	42	-0.12	-0.618	80
8	**N45**	1350	13500	860	10800	955	12000	350	44	-0.12	-0.618	80
9	**N48**	1400	14000	836	10500	875	11000	374	47	-0.12	-0.618	80
10	**N50**	1425	14250	836	10500	875	11000	390	49	-0.12	-0.618	80
11	**N52**	1450	14500	836	10500	875	11000	406	51	-0.12	-0.618	60
12	**N55**	1490	14900	716	9000	876	11000	430	54	-0.15	-0.618	60
13	**N33M**	1175	11750	836	10500	1114	14000	267	34	-0.12	-0.595	100
14	**N35M**	1210	12100	868	10900	1114	14000	282.5	35	-0.12	-0.595	100
15	**N38M**	1260	12600	899	11300	1114	14000	306.5	39	-0.12	-0.595	100
16	**N40M**	1270	12700	923	11600	1114	14000	322	40	-0.12	-0.595	100
17	**N42M**	1315	13150	955	12000	1114	14000	338	42	-0.12	-0.595	100
18	**N45M**	1350	13500	971	12200	1114	14000	354	44	-0.12	-0.595	100
19	**N48M**	1395	13950	995	12500	1114	14000	378	48	-0.12	-0.595	100
20	**N30H**	1105	11050	796	10000	1353	17000	235	30	-0.12	-0.572	120
21	**N33H**	1175	11750	836	10500	1353	17000	267	34	-0.12	-0.572	120
22	**N35H**	1210	12100	868	10900	1353	17000	282.5	35	-0.12	-0.572	120
23	**N38H**	1260	12600	899	11300	1353	17000	306.5	39	-0.12	-0.572	120
24	**N40H**	1270	12700	923	11600	1353	17000	322	40	-0.12	-0.572	120
25	**N42H**	1300	13000	955	12000	1353	17000	330	41	-0.12	-0.572	120
26	**N45H**	1350	13500	971	12200	1353	17000	354	44	-0.12	-0.572	120
27	**N48H**	1390	13900	1011	12700	1273	16000	378	48	-0.12	-0.572	120
28	**N50H**	1425	14250	836	10500	1273	16000	390	49	-0.12	-0.618	140
29	**N30SH**	1125	11250	811	10200	1592	20000	243	31	-0.12	-0.549	150
30	**N33SH**	1175	11750	844	10600	1592	20000	267	34	-0.12	-0.549	150
31	**N35SH**	1210	12100	876	11000	1592	20000	282.5	35	-0.12	-0.549	150
32	**N38SH**	1260	12600	907	11400	1592	20000	306.5	39	-0.12	-0.549	150
33	**N40SH**	1270	12700	939	11800	1592	20000	322	40	-0.12	-0.549	150
34	**N42SH**	1310	13100	955	12000	1592	20000	330	41	-0.12	-0.549	150
35	**N45SH**	1350	13500	979	12300	1592	20000	354	44	-0.12	-0.549	150
36	**N48SH**	1390	13900	995	12500	1512	19000	378	48	-0.12	-0.572	150
37	**N28UH**	1075	10750	764	9600	1990	25000	227	29	-0.12	-0.51	180
38	**N30UH**	1125	11250	812	10200	1990	25000	243	31	-0.12	-0.51	180
39	**N33UH**	1175	11750	852	10700	1990	25000	267	34	-0.12	-0.51	180
40	**N35UH**	1210	12100	860	10800	1990	25000	282.5	35	-0.12	-0.51	180
41	**N38UH**	1260	12600	876	11000	1990	25000	306.5	39	-0.12	-0.51	180
42	**N40UH**	1270	12700	915	11500	1990	25000	318	40	-0.12	-0.51	180

Click for: Arnold Neodymium Iron Boron Webpage

Click for: Grain Boundary Diffused Neo Catalog

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770 Linden Avenue, Rochester, NY 14625
Ph: (+1) 585-385-9010
E-mail: info@arnoldmagnetics.com
www.arnoldmagnetics.com

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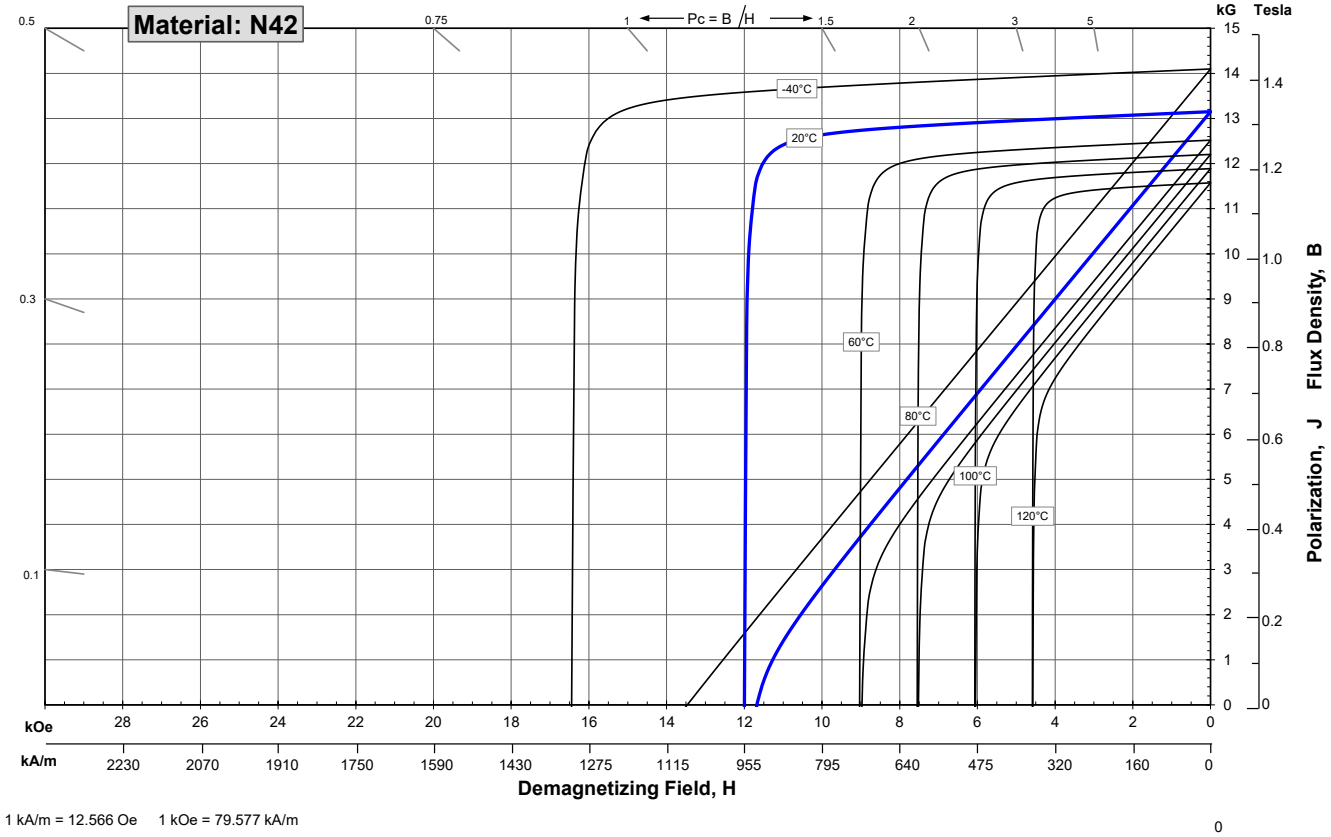
Sintered Neodymium-Iron-Boron Magnets

These are also referred to as "Neo" or NdFeB magnets. They offer a combination of high magnetic output at moderate cost. Please contact Arnold for additional grade information and recommendations for protective coating. Assemblies using these magnets can also be provided.

	Characteristic	Units	min.	nominal	max.
Magnetic Properties	**Br**, Residual Induction	Gauss	12,800	13,150	13,500
		mT	1280	1315	1350
	H_{cB}, Coercivity	Oersteds	10,800	11,850	12,900
		kA/m	860	943	1027
	H_{cJ}, Intrinsic Coercivity	Oersteds	12,000		
		kA/m	955		
	BHmax, Maximum Energy Product	MGOe	40	42	44
		kJ/m³	318	334	350

	Characteristic	Units	C //	C ⊥
Thermal Properties	Reversible Temperature Coefficients (1)			
	of Induction, α(Br)	%/ºC	-0.12	
	of Coercivity, α(Hcj)	%/ºC	-0.62	
	Coefficient of Thermal Expansion (2)	ΔL/L per ºCx10⁻⁶	7	-1
	Thermal Conductivity	kcal/mhrºC	5.3	5.8
	Specific Heat (3)	cal/gºC	0.11	
	Curie Temperature, Tc	ºC	310	
Other Properties	Flexural Strength	psi	41,300	
		MPa	285	
	Density	g/cm³	7.6	
	Hardness, Vickers	Hv	620	
	Electrical Resistivity, ρ	μΩ • cm	150 // 130 ⊥	

Notes: (1) Coefficients measured between 20 and 80 ºC
(2) Between 20 and 200 ºC. Values are typical and can vary.
(3) Between 20 and 140 ºC



1 kA/m = 12.566 Oe 1 kOe = 79.577 kA/m

Notes The material data and demagnetization curves shown above represent typical properties that may vary due to product shape and size.
Demagnetization curves show nominal Br and minimum Hci.
Magnets can be supplied thermally stabilized or magnetically calibrated to customer specifications.
Additional grades are available. Please contact the factory for information.

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ARNOLD® MAGNETIC TECHNOLOGIES

N52

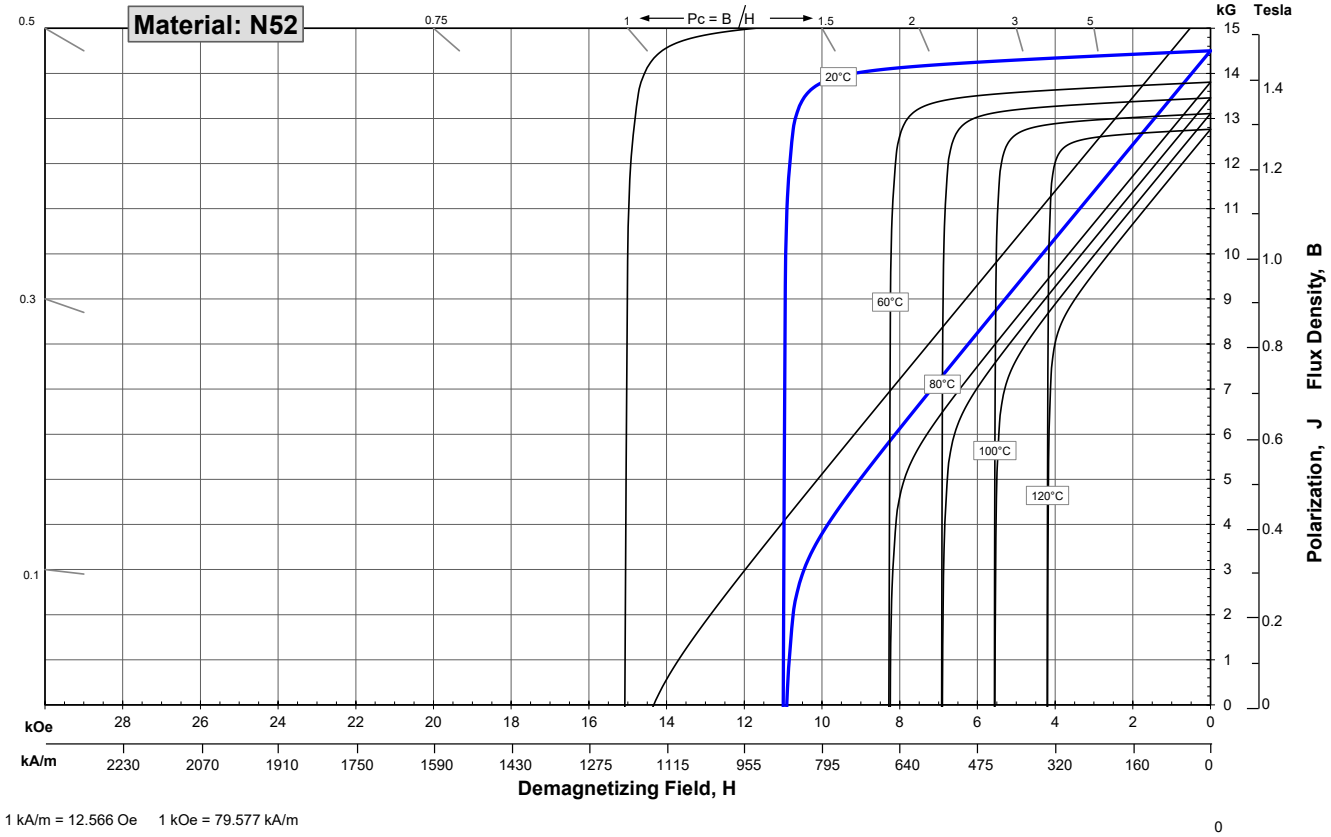
Sintered Neodymium-Iron-Boron Magnets

These are also referred to as "Neo" or NdFeB magnets. They offer a combination of high magnetic output at moderate cost. Please contact Arnold for additional grade information and recommendations for protective coating. Assemblies using these magnets can also be provided.

	Characteristic	Units	min.	nominal	max.
Magnetic Properties	**Br**, Residual Induction	Gauss	14,200	14,500	14,800
		mT	1420	1450	1480
	H_{cB}, Coercivity	Oersteds	10,500	12,300	14,100
		kA/m	836	979	1122
	H_{cJ}, Intrinsic Coercivity	Oersteds	11,000		
		kA/m	875		
	BHmax, Maximum Energy Product	MGOe	49	51	53
		kJ/m³	390	406	422

	Characteristic	Units	C //	C ⊥
Thermal Properties	Reversible Temperature Coefficients (1)			
	of Induction, α(Br)	%/ºC	-0.12	
	of Coercivity, α(Hcj)	%/ºC	-0.62	
	Coefficient of Thermal Expansion (2)	ΔL/L per ºCx10⁻⁶	7	-1
	Thermal Conductivity	kcal/mhrºC	5.3	5.8
	Specific Heat (3)	cal/gºC	0.11	
	Curie Temperature, Tc	ºC	310	
Other Properties	Flexural Strength	psi	41,300	
		MPa	285	
	Density	g/cm³	7.6	
	Hardness, Vickers	Hv	620	
	Electrical Resistivity, ρ	μΩ • cm	150 // 130 ⊥	

Notes: (1) Coefficients measured between 20 and 60 ºC
(2) Between 20 and 200 ºC. Values are typical and can vary.
(3) Between 20 and 140 ºC



1 kA/m = 12.566 Oe 1 kOe = 79.577 kA/m

Notes The material data and demagnetization curves shown above represent typical properties that may vary due to product shape and size.
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ARNOLD® MAGNETIC TECHNOLOGIES

N30H

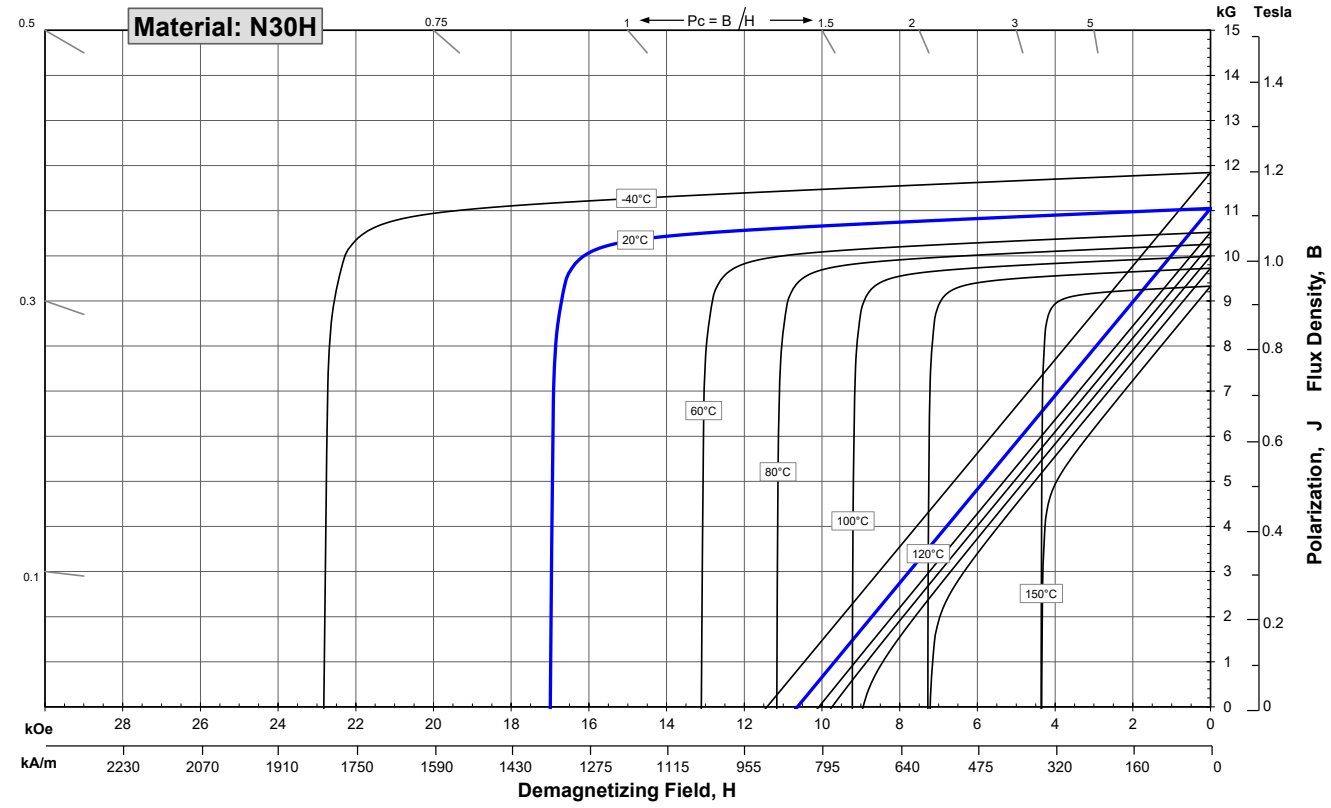
Sintered Neodymium-Iron-Boron Magnets

These are also referred to as "Neo" or NdFeB magnets. They offer a combination of high magnetic output at moderate cost. Please contact Arnold for additional grade information and recommendations for protective coating. Assemblies using these magnets can also be provided.

	Characteristic	Units	min.	nominal	max.
Magnetic Properties	**Br**, Residual Induction	Gauss	10,800	11,050	11,300
		mT	1080	1105	1130
	H_{cB}, Coercivity	Oersteds	10,000	10,400	10,800
		kA/m	796	828	859
	H_{cJ}, Intrinsic Coercivity	Oersteds	17,000		
		kA/m	1,353		
	BHmax, Maximum Energy Product	MGOe	28	30	31
		kJ/m^3	223	235	247

	Characteristic	Units	C //	C ⊥
Thermal Properties	Reversible Temperature Coefficients (1)			
	of Induction, α(Br)	%/ºC	-0.12	
	of Coercivity, α(Hcj)	%/ºC	-0.57	
	Coefficient of Thermal Expansion (2)	ΔL/L per ºCx10^{-6}	7	-1
	Thermal Conductivity	kcal/mhrºC	5.3	5.8
	Specific Heat (3)	cal/gºC	0.11	
	Curie Temperature, Tc	ºC	310	
Other Properties	Flexural Strength	psi	41,300	
		MPa	285	
	Density	g/cm^3	7.6	
	Hardness, Vickers	Hv	620	
	Electrical Resistivity, ρ	μΩ • cm	150 // 130 ⊥	

Notes: (1) Coefficients measured between 20 and 120 ºC
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(3) Between 20 and 140 ºC



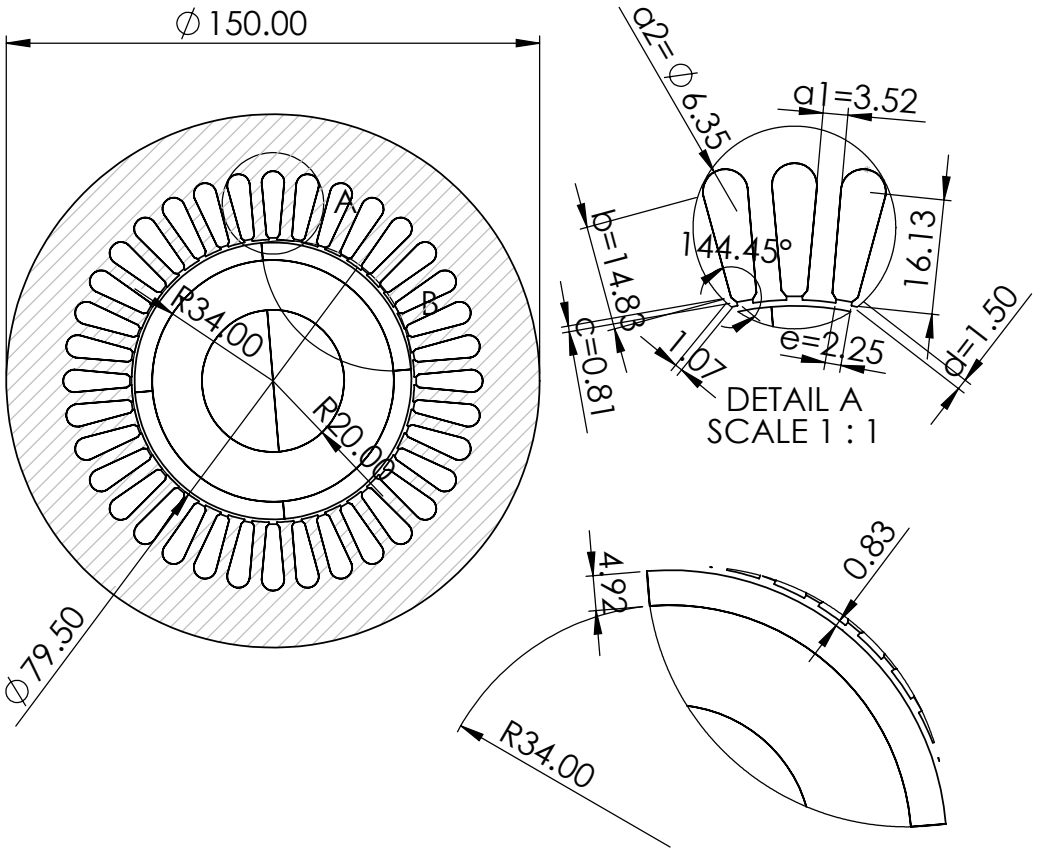
1 kA/m = 12.566 Oe 1 kOe = 79.577 kA/m

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Appendix C

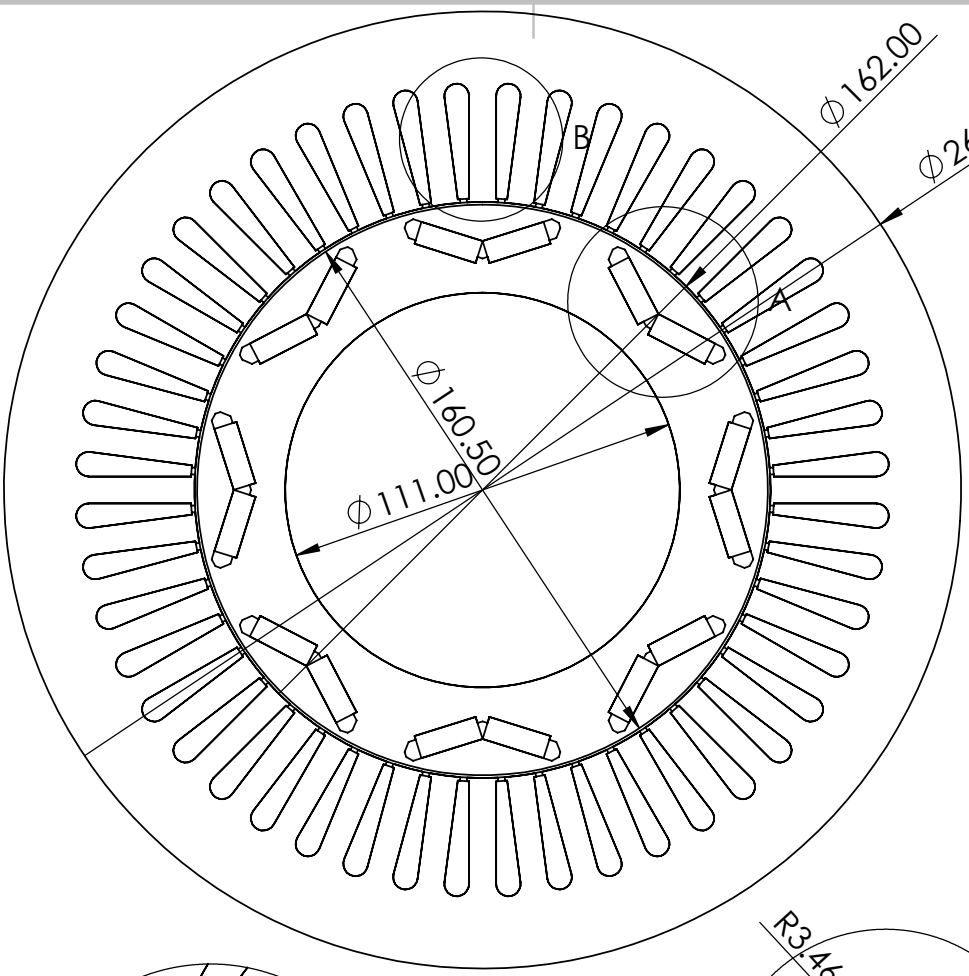
2D drawings

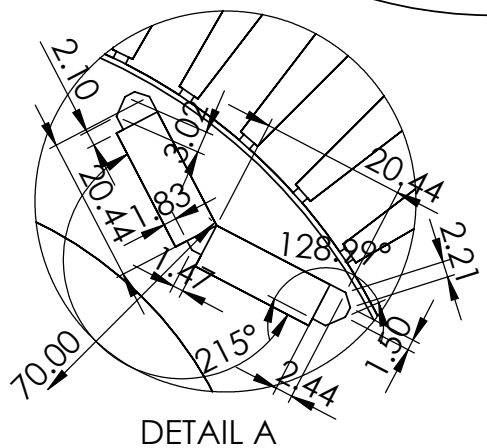


ITEM NO.	PART NUMBER	Material	QTY.
1	Stator	AISI STEEL 1010	1
2	coil	Copper	36
3	Rotorf	AISI STELL 1010	1
4	Band^Assem1	Virtual component	1
5	Air Cylinder	AIR	1

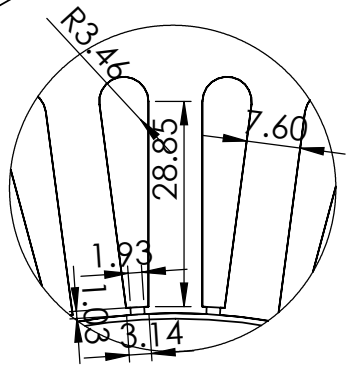
SCALE:1/2 Aymen Bahri PMSM

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DETAIL A
SCALE 1 : 1

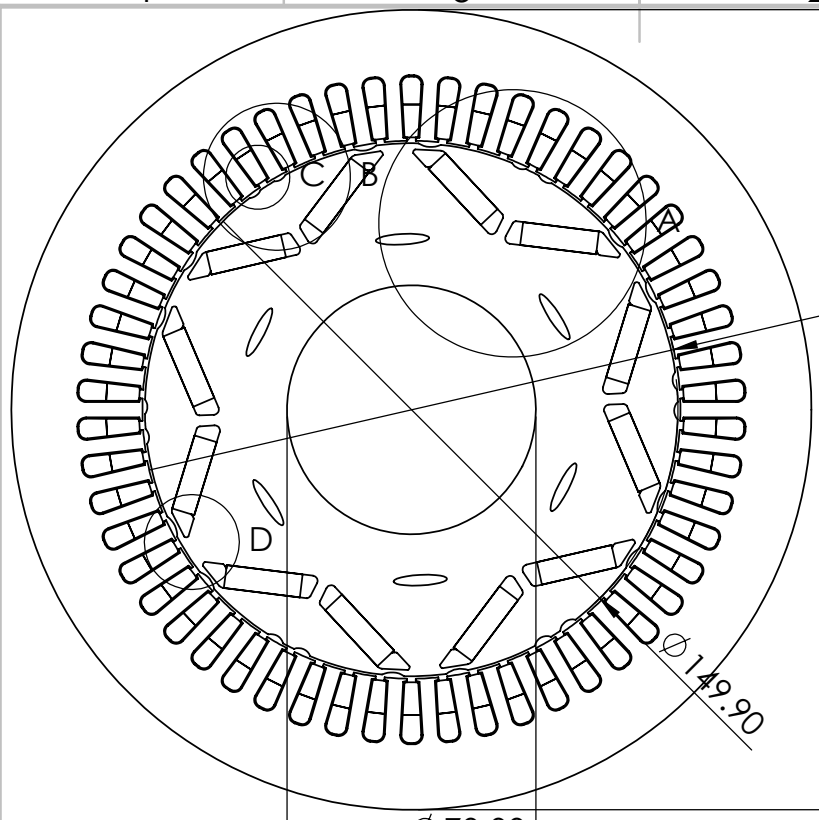


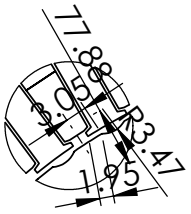
DETAIL B
SCALE 1 : 1

ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	stator components		1
2	Rotor Components		1
3	Air Circle^Toyota		1
4	Part1^Toyota		1

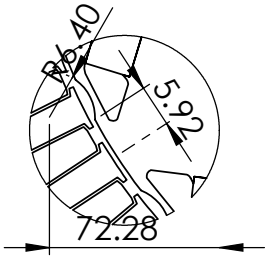
SCALE:1/2 Aymen Bahri Toyota Prius 2004
IPMSM Drawing

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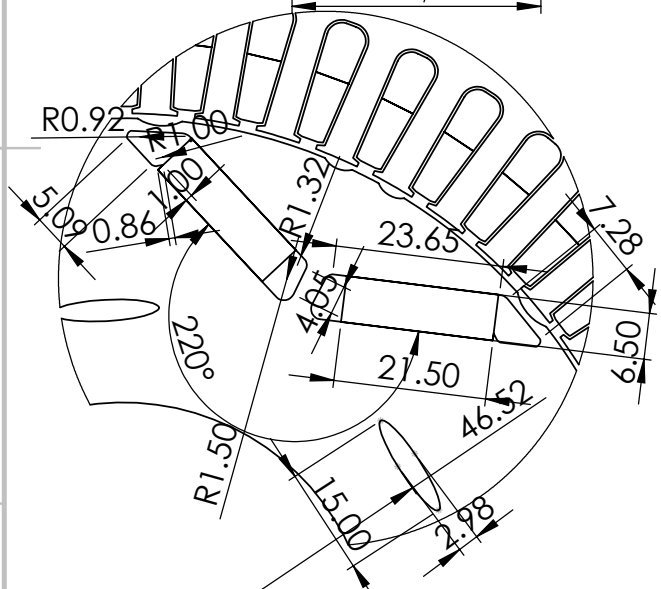




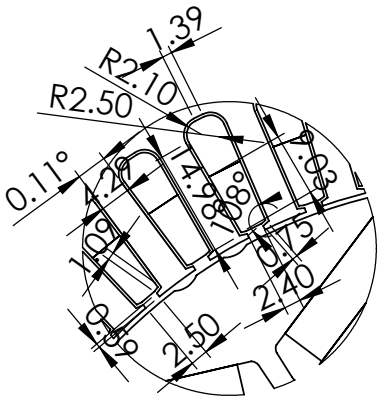
DETAIL C
SCALE 1 : 1



DETAIL D
SCALE 1 : 1



DETAIL A
SCALE 1 : 1



DETAIL B
SCALE 1 : 1

ITEM NO.	PART NUMBER	MATERIAL	QTY.
1	stator	M-27	1
2	coil_1	COPPER	54
3	coil_2	COPPER	54
4	RotorF	N52 AND M-27	1
5	Band^TM312	VIRTUAL COMPONENT	1
6	Air Circle^TM312	AIR	1

SCALE:1/2

Aymen Bahri

Tesla Model 3
IPMSM drawing

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